EMLN analysis

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Data

Here I read the interaction data as data.frames imported from csv or tab-separated files. Then I extract just the interaction matrices with the corresponding row names.

```
pm.disp<- read.table(here::here("data/raw/multiplex/pmah_disp.txt"), sep="\t",
                     dec=".", header= TRUE)
pm.poll<- read.table(here::here("data/raw/multiplex/pmah_poll.txt"),
    na="NA", sep="\t", dec=".", header= TRUE)
pm.herb<- read.table(here::here("data/raw/multiplex/pmah_herb.txt"),
     na="NA", sep="\t", dec=".", header= TRUE)
pm.attr<- read.table(here::here("data/raw/multiplex/pmah_attr.txt"),
    na="NA", sep="\t", dec=".", header= TRUE)

# Interaction matrices extracted from the dataframes.
pm.herb.mat<- as.matrix(pm.herb[1:19,2:12],
        ncol = n, dimnames = list(paste0("A", 1:12), paste0("P", 1:19)))
row.names(pm.herb.mat)<- pm.herb$X[1:19]

pm.poll.mat<- as.matrix(pm.poll[1:19,2:44],
        dimnames = list(paste0("A", 1:44), paste0("P", 1:19)))
row.names(pm.poll.mat)<- pm.poll$X[1:19]

pm.disp.mat<- as.matrix(pm.disp[1:19,2:22],
        dimnames = list(paste0("A", 1:22), paste0("P", 1:19)))
row.names(pm.disp.mat)<- pm.disp$X[1:19]

# Total number of distinct nodes (physical nodes).
# Here the trees are counted just once.
dim(pm.herb.mat)[1]+dim(pm.herb.mat)[2]+dim(pm.poll.mat)[2]+dim(pm.disp.mat)[2]
```

```
[1] 94
```

```
# Total number of state nodes (trees). Here the trees are counted just once.
dim(pm.herb.mat)[1]
```

```
[1] 19
```

```
# Total number of nodes (physical and state).
# Here the trees (state nodes) are counted three times, one state in each layer.
dim(pm.herb.mat)[1]+dim(pm.herb.mat)[2]+dim(pm.poll.mat)[1]+dim(pm.poll.mat)[2]+
    dim(pm.disp.mat)[1]+dim(pm.disp.mat)[2]
```

```
[1] 132
```

```r
# Total number of interactions recorded
dim(pm.herb.mat); dim(pm.poll.mat); dim(pm.disp.mat)
```

```
[1] 19 11
[1] 19 43
[1] 19 21
```

```r
sum(pm.herb.mat, pm.poll.mat, pm.disp.mat)
```

```
[1] 5427
```

```r
non_zero_count <- sum(pm.herb.mat != 0,   # Count non-zero elements.
                                          # No. distinct interactions.
                      pm.poll.mat != 0,
                      pm.disp.mat != 0) ; non_zero_count
```

```
[1] 493
```

```r
# Convert a matrix to an edge list for bipartite network
pm.disp.bip<- matrix_to_list_bipartite(pm.disp.mat,
                                       group_names = c('A_set',
                                                       'P_set'))$edge_list

pm.poll.bip<- matrix_to_list_bipartite(pm.poll.mat,
                                       group_names = c('A_set',
                                                       'P_set'))$edge_list

pm.herb.bip<- matrix_to_list_bipartite(pm.herb.mat,
                                       group_names = c('A_set',
                                                       'P_set'))$edge_list
```

Multilayer structure

This is code to create the multilayer structure.

```r
# Create the layers from the matrix. These are  the intra-layer interactions,
# based on the adjacency matrices.
herbivory_layer <- pm.herb.mat
pollination_layer <- pm.poll.mat
seed_dispersal_layer <- pm.disp.mat

layer_attrib <- tibble(layer_id=1:3,
                       layer_name=c('Herbivory','Pollination',
                                    'Seed dispersal'),
                       Method=c('Observation','Collection','Observation'))

# Create the ELL tibble with interlayer links.
interlayer <- tibble(layer_from= c(rep('Herbivory',19),
                                   rep('Pollination',19)),
                     node_from= c("pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865",
                                 "pm1933","pm1931","pm1932","pm1934","pm1936",
```

```
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865"),
                     layer_to= c(rep('Pollination',19),
                                 rep('Seed dispersal',19)),
                     node_to= c("pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865",
                                 "pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm1939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865"),
                                 weight= 1) # I'm assuming here
                                            # interlayer links weights= 1.

# Create the multilayer. It has 132 nodes, counting the trees just as physical
# nodes (only once).

multilayer_bip_pru <- create_multilayer_network(list_of_layers =
                       list(herbivory_layer, pollination_layer,
                            seed_dispersal_layer),
                       bipartite = T,
                       directed = F,
                       interlayer_links = interlayer,
                       layer_attributes = layer_attrib)
```

```
[1] "Layer #1 processing."
[1] "Input: a bipartite matrix"
[1] "Done."
[1] "Layer #2 processing."
[1] "Input: a bipartite matrix"
[1] "Done."
[1] "Layer #3 processing."
[1] "Input: a bipartite matrix"
[1] "Done."
[1] "Creating extended link list with node IDs"
[1] "Organizing state nodes"
```

emln Analysis. SAM matrix

Here I get the supra-adjacency matrix (SAM) and generate a the whole graph.

```
library(emln)
library(tidyverse)
library(infomapecology)
```

```
Loading required package: attempt


Attaching package: 'attempt'

The following objects are masked from 'package:dplyr':

    if_all, if_any, if_else
```

```
Loading required package: cowplot

Attaching package: 'cowplot'

The following object is masked from 'package:ggpubr':

    get_legend

The following object is masked from 'package:lubridate':

    stamp

Loading required package: ggalluvial

Loading required package: rlang

Attaching package: 'rlang'

The following objects are masked from 'package:statnet.common':

    base_env, empty_env

The following objects are masked from 'package:hablar':

    chr, dbl, int, lgl

The following object is masked from 'package:igraph':

    is_named

The following object is masked from 'package:magrittr':

    set_names

The following objects are masked from 'package:purrr':

    %@%, flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```

```r
library(magrittr)

# Get the SAM. Remove zero rows/cols and get a sparse matrix. This has 279 nodes,
# including the trees counted three times and animals counted three times. The full
# list can found in multilayer_bip_pru$nodes and the complete list of nodes
# (including the NAs of the large megamatrix, in: multilayer_bip_pru$state_nodes_map.
pru.sam_bipartite <- get_sam(multilayer = multilayer_bip_pru, bipartite = T,
                             directed = F, sparse = F, remove_zero_rows_cols = T)
```

```
Joining with `by = join_by(node_name, layer_name)`
```

```r
dim(pru.sam_bipartite$M) # This is a 132x132 square supra-adjacency matrix.
```

```
[1] 132 132
```

```r
pru.sam_bipartite$M
```

```
     1 31 37 38 40 44 45 46 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70
1    0  0  0  0  0  0  0  0  0  0  0  0 10  5  0  0  0  0  5  0  0  0 11  0  0
31   0  0  0  0  0  0  0  0  0  4 11  0 30  3 17  0  0 10  5  8  0  0 30  0  0
```

37	0	0	0	0	0	0	0	0	0	30	10	8	8	41	59	8	6	0	18	11	5	18	4	23	6
38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	62	0	0	0	0	0
40	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	10	0	0	5	0	0
44	0	0	0	0	0	0	0	0	20	0	0	0	0	0	10	0	0	0	0	5	0	0	0	0	5
45	0	0	0	0	0	0	0	0	1	3	0	0	1	0	2	1	0	0	2	0	0	0	0	1	0
46	0	0	0	0	0	0	0	0	2	0	0	0	0	5	0	0	0	0	8	0	0	0	0	0	0
54	0	0	0	0	0	20	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
55	0	4	30	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
56	0	11	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
57	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
58	10	30	8	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
59	5	3	41	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
60	0	17	59	0	0	10	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
61	0	0	8	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
62	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
63	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
64	5	5	18	0	18	0	2	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
65	0	8	11	62	10	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
66	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
67	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
68	11	30	4	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
69	0	0	23	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
70	0	0	6	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
71	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
72	0	0	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
86	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	5	0	2	0	0	0
89	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	60	10	0	0	0	60	0	0
93	0	0	0	0	0	0	0	0	20	0	0	0	15	8	0	0	0	5	0	0	0	0	0	0	10
95	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
96	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
97	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
98	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
99	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
101	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
102	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
103	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
104	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
105	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
106	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
107	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
108	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
109	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
110	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
112	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
114	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
115	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
117	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
118	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
119	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
120	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
121	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

126	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
127	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
128	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
132	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
133	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
134	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
135	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
136	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
140	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
147	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
148	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
149	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
150	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
151	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
152	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
153	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
154	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
155	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
156	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
157	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
158	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
159	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
160	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
161	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
162	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
163	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
164	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
165	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
166	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
167	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
169	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
176	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
177	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
178	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
184	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
185	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
206	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
209	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
215	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
216	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
218	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
234	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
235	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
236	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
237	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
238	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
239	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
240	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
241	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
242	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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63	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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66	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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68	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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86	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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126	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
127	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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147	22	1	0	0	0	3	0	0	6	1	0	0	0	0	0	0	0	0	0
148	15	1	1	0	0	9	0	0	14	11	0	0	0	0	0	0	0	0	0
149	3	7	0	0	0	0	3	3	57	2	1	0	0	0	0	0	0	0	0
150	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
151	29	0	0	0	0	11	0	0	1	22	0	0	0	1	0	0	0	0	0
152	8	0	0	0	0	0	0	0	0	2	1	0	0	0	0	1	0	0	0
153	16	0	0	0	0	2	2	0	6	1	2	2	0	0	0	0	0	0	2
154	98	7	1	1	3	41	6	2	10	21	0	0	0	0	0	0	4	0	0
155	8	0	0	0	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0
156	10	0	0	0	0	1	0	0	1	0	1	2	0	0	0	0	0	0	0
157	32	2	0	2	0	8	0	1	6	8	0	0	0	0	1	0	1	0	0
158	8	0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0
159	49	2	0	0	0	0	0	0	3	1	1	0	0	0	0	0	0	0	0
160	66	6	0	1	2	32	3	1	8	50	0	0	0	0	0	0	0	0	0
161	166	0	0	0	0	1	3	0	0	2	1	1	1	0	0	0	0	1	4
162	3	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
163	23	0	1	0	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0
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274	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
31	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
37	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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54	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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56	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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61	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
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68	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
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70	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
71	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
72	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

86	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
89	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
93	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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96	0	0	1	0	0	1	2	0	0	1	0	0	0	0	1	0	0	0	0
97	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0
98	0	0	0	0	0	5	18	0	8	0	0	0	0	0	14	7	0	0	0
99	5	1	56	43	3	32	112	2	2	28	2	34	3	0	96	13	3	45	2
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102	0	0	0	0	0	1	2	0	0	0	0	0	0	0	1	0	0	0	0
103	0	2	1	0	1	1	4	0	1	0	0	1	0	2	3	0	1	2	0
104	2	7	0	0	0	0	0	15	0	0	10	0	0	9	0	0	4	0	9
105	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0
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111	22	15	3	2	29	8	16	98	8	10	32	8	49	66	166	3	23	1	18
112	1	1	7	0	0	0	0	7	0	0	2	0	2	6	0	0	0	0	0
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117	0	0	0	0	0	0	0	3	0	0	0	0	0	2	0	0	0	0	0
118	3	9	0	0	11	0	2	41	0	1	8	0	0	32	1	0	2	0	3
119	0	0	3	0	0	0	2	6	0	0	0	0	0	3	3	0	0	0	0
120	0	0	3	0	0	0	0	2	0	0	1	0	0	1	0	0	0	0	0
121	6	14	57	0	1	0	6	10	0	1	6	0	3	8	0	0	0	0	0
126	1	11	2	0	22	2	1	21	2	0	8	2	1	50	2	0	2	0	6
127	0	0	1	0	0	1	2	0	0	1	0	0	1	0	1	0	0	2	0
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140	0	0	0	0	0	0	2	0	0	0	0	0	0	0	4	0	0	0	0
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176	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0
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56	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
60	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
61	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
62	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
63	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
64	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
65	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
66	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
67	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
68	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
69	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
70	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
71	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

72	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
86	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
89	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
93	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
95	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
96	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
97	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
98	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
99	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
101	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
102	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
103	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
104	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
105	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
106	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
107	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
108	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
109	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
110	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
112	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
114	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
115	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
117	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
118	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
119	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
120	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
121	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
126	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
127	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
128	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
132	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
133	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
134	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
135	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
136	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
140	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
147	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
148	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
149	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
150	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
151	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
152	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
153	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
154	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
155	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
156	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
157	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
158	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
159	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
160	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

161	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
162	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
163	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
164	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
165	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
166	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
167	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
169	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
176	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
177	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
178	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
184	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
185	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
206	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0
209	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
215	1	15	9	3	0	0	3	0	5	0	2	6	12	16	26	14	7	78	3
216	39	147	37	87	28	1	51	34	48	6	1	15	10	31	39	47	24	3	14
218	7	20	1	4	2	0	2	2	4	1	0	0	0	0	0	6	8	4	0
234	2	59	6	10	0	0	10	21	20	8	0	4	4	5	3	21	33	5	41
235	0	25	8	1	2	0	2	0	2	0	0	12	8	5	7	33	2	1	24
236	0	0	0	2	0	0	3	0	0	2	0	0	0	1	1	1	1	0	0
237	0	7	0	1	0	0	4	3	3	0	0	0	4	0	0	4	0	0	0
238	26	27	31	28	65	9	64	33	20	8	5	20	13	4	14	134	8	32	47
239	1	3	3	2	42	2	13	6	2	2	0	7	5	0	1	13	3	0	7
240	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
241	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
242	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
243	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
244	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
245	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
246	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
247	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
248	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
249	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
250	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
251	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
252	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
253	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
254	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
255	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
256	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
257	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
258	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
261	0	9	0	5	0	0	0	0	0	0	0	0	0	1	0	0	2	1	7
263	0	2	1	0	1	0	2	0	7	0	0	1	3	1	6	0	1	0	0
264	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	1	1	1	2
265	0	0	0	0	0	0	1	1	0	0	1	3	4	1	0	27	2	1	1
266	0	0	0	0	3	5	11	2	7	0	3	25	27	6	3	32	5	8	13
267	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0
268	0	0	0	0	0	0	0	1	2	0	4	3	7	1	4	0	0	0	0
273	1	11	16	1	3	0	1	3	6	0	2	20	8	6	4	4	5	25	5
274	10	168	66	13	0	0	2	1	18	0	0	0	1	0	0	24	34	1	9
276	0	1	2	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0

	261	263	264	265	266	267	268	273	274	276
1	0	0	0	0	0	0	0	0	0	0
31	0	0	0	0	0	0	0	0	0	0
37	0	0	0	0	0	0	0	0	0	0
38	0	0	0	0	0	0	0	0	0	0
40	0	0	0	0	0	0	0	0	0	0
44	0	0	0	0	0	0	0	0	0	0
45	0	0	0	0	0	0	0	0	0	0
46	0	0	0	0	0	0	0	0	0	0
54	0	0	0	0	0	0	0	0	0	0
55	0	0	0	0	0	0	0	0	0	0
56	0	0	0	0	0	0	0	0	0	0
57	0	0	0	0	0	0	0	0	0	0
58	0	0	0	0	0	0	0	0	0	0
59	0	0	0	0	0	0	0	0	0	0
60	0	0	0	0	0	0	0	0	0	0
61	0	0	0	0	0	0	0	0	0	0
62	0	0	0	0	0	0	0	0	0	0
63	0	0	0	0	0	0	0	0	0	0
64	0	0	0	0	0	0	0	0	0	0
65	0	0	0	0	0	0	0	0	0	0
66	0	0	0	0	0	0	0	0	0	0
67	0	0	0	0	0	0	0	0	0	0
68	0	0	0	0	0	0	0	0	0	0
69	0	0	0	0	0	0	0	0	0	0
70	0	0	0	0	0	0	0	0	0	0
71	0	0	0	0	0	0	0	0	0	0
72	0	0	0	0	0	0	0	0	0	0
86	0	0	0	0	0	0	0	0	0	0
89	0	0	0	0	0	0	0	0	0	0
93	0	0	0	0	0	0	0	0	0	0
95	0	0	0	0	0	0	0	0	0	0
96	0	0	0	0	0	0	0	0	0	0
97	0	0	0	0	0	0	0	0	0	0
98	0	0	0	0	0	0	0	0	0	0
99	0	0	0	0	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0	0	0
101	0	0	0	0	0	0	0	0	0	0
102	0	0	0	0	0	0	0	0	0	0
103	0	0	0	0	0	0	0	0	0	0
104	0	0	0	0	0	0	0	0	0	0
105	0	0	0	0	0	0	0	0	0	0
106	0	0	0	0	0	0	0	0	0	0
107	0	0	0	0	0	0	0	0	0	0
108	0	0	0	0	0	0	0	0	0	0
109	0	0	0	0	0	0	0	0	0	0
110	0	0	0	0	0	0	0	0	0	0
111	0	0	0	0	0	0	0	0	0	0
112	0	0	0	0	0	0	0	0	0	0
114	0	0	0	0	0	0	0	0	0	0
115	0	0	0	0	0	0	0	0	0	0
117	0	0	0	0	0	0	0	0	0	0
118	0	0	0	0	0	0	0	0	0	0

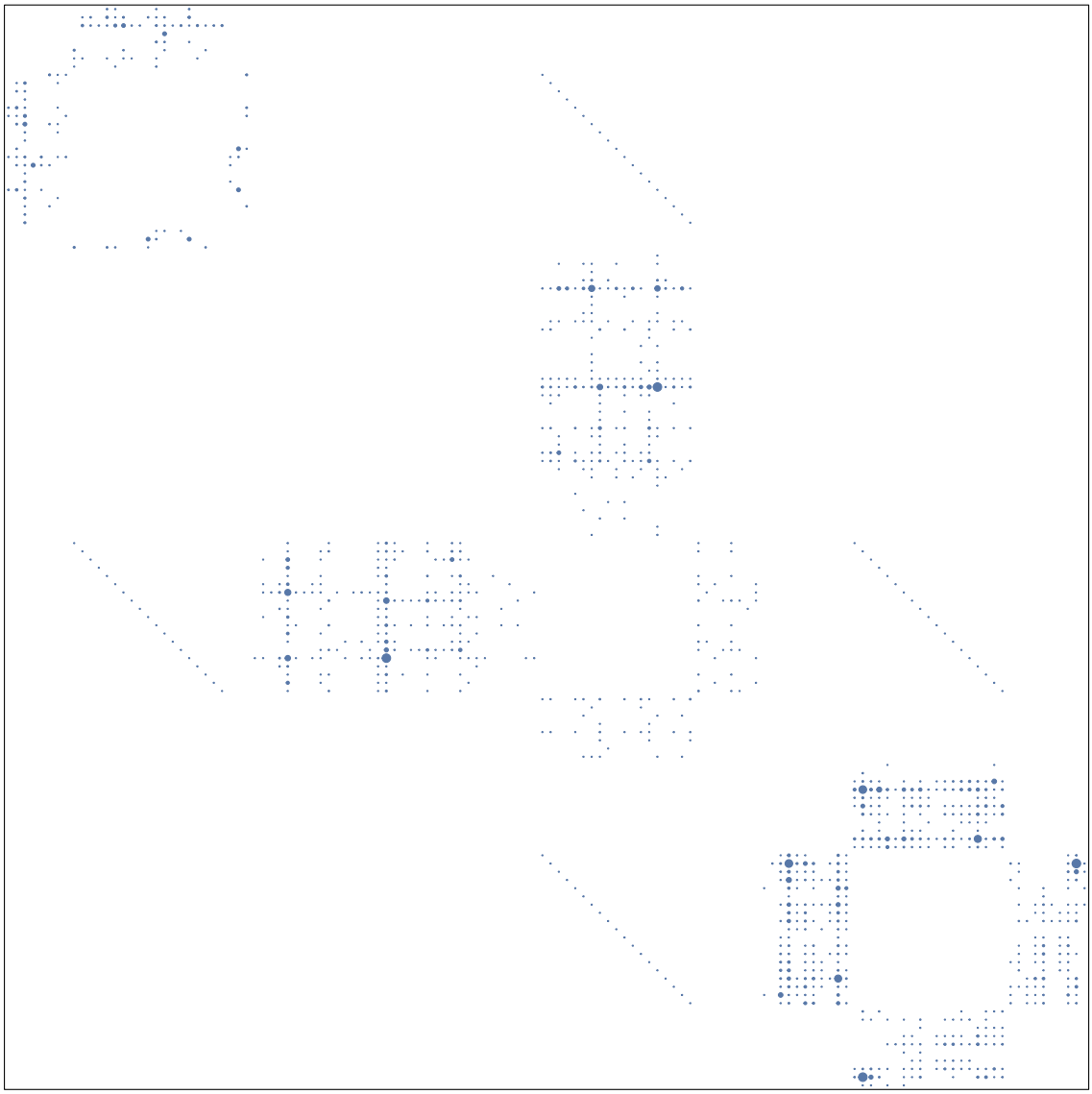
119	0	0	0	0	0	0	0	0	0	0
120	0	0	0	0	0	0	0	0	0	0
121	0	0	0	0	0	0	0	0	0	0
126	0	0	0	0	0	0	0	0	0	0
127	0	0	0	0	0	0	0	0	0	0
128	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0
132	0	0	0	0	0	0	0	0	0	0
133	0	0	0	0	0	0	0	0	0	0
134	0	0	0	0	0	0	0	0	0	0
135	0	0	0	0	0	0	0	0	0	0
136	0	0	0	0	0	0	0	0	0	0
140	0	0	0	0	0	0	0	0	0	0
147	0	0	0	0	0	0	0	0	0	0
148	0	0	0	0	0	0	0	0	0	0
149	0	0	0	0	0	0	0	0	0	0
150	0	0	0	0	0	0	0	0	0	0
151	0	0	0	0	0	0	0	0	0	0
152	0	0	0	0	0	0	0	0	0	0
153	0	0	0	0	0	0	0	0	0	0
154	0	0	0	0	0	0	0	0	0	0
155	0	0	0	0	0	0	0	0	0	0
156	0	0	0	0	0	0	0	0	0	0
157	0	0	0	0	0	0	0	0	0	0
158	0	0	0	0	0	0	0	0	0	0
159	0	0	0	0	0	0	0	0	0	0
160	0	0	0	0	0	0	0	0	0	0
161	0	0	0	0	0	0	0	0	0	0
162	0	0	0	0	0	0	0	0	0	0
163	0	0	0	0	0	0	0	0	0	0
164	0	0	0	0	0	0	0	0	0	0
165	0	0	0	0	0	0	0	0	0	0
166	0	0	0	0	0	0	0	0	0	0
167	0	0	0	0	0	0	0	0	0	0
169	0	0	0	0	0	0	0	0	0	0
176	0	0	0	0	0	0	0	0	0	0
177	0	0	0	0	0	0	0	0	0	0
178	0	0	0	0	0	0	0	0	0	0
184	0	0	0	0	0	0	0	0	0	0
185	0	0	0	0	0	0	0	0	0	0
206	0	0	0	0	0	0	0	0	0	0
209	0	0	0	0	0	0	0	0	0	0
215	0	0	0	0	0	0	0	0	0	0
216	0	0	0	0	0	0	0	0	0	0
218	0	0	0	0	0	0	0	0	0	0
234	0	0	0	0	0	0	0	0	0	0
235	0	0	0	0	0	0	0	0	0	0
236	0	0	0	0	0	0	0	0	0	0
237	0	0	0	0	0	0	0	0	0	0
238	0	0	0	0	0	0	0	0	0	0
239	0	0	0	0	0	0	0	0	0	0
240	0	0	0	0	0	0	0	1	10	0
241	9	2	0	0	0	0	0	11	168	1

```
242   0   1   0   0   0   0   0  16  66   2
243   5   0   0   0   0   0   0   1  13   0
244   0   1   0   0   3   0   0   3   0   2
245   0   0   0   0   5   0   0   0   0   0
246   0   2   0   1  11   2   0   1   2   1
247   0   0   0   1   2   0   1   3   1   0
248   0   7   3   0   7   2   2   6  18   0
249   0   0   0   0   0   0   0   0   0   0
250   0   0   0   1   3   0   4   2   0   0
251   0   1   0   3  25   0   3  20   0   0
252   0   3   0   4  27   0   7   8   1   0
253   1   1   0   1   6   0   1   6   0   0
254   0   6   0   0   3   0   4   4   0   0
255   0   0   1  27  32   0   0   4  24   0
256   2   1   1   2   5   0   0   5  34   0
257   1   0   1   1   8   0   0  25   1   0
258   7   0   2   1  13   0   0   5   9   0
261   0   0   0   0   0   0   0   0   0   0
263   0   0   0   0   0   0   0   0   0   0
264   0   0   0   0   0   0   0   0   0   0
265   0   0   0   0   0   0   0   0   0   0
266   0   0   0   0   0   0   0   0   0   0
267   0   0   0   0   0   0   0   0   0   0
268   0   0   0   0   0   0   0   0   0   0
273   0   0   0   0   0   0   0   0   0   0
274   0   0   0   0   0   0   0   0   0   0
276   0   0   0   0   0   0   0   0   0   0
```

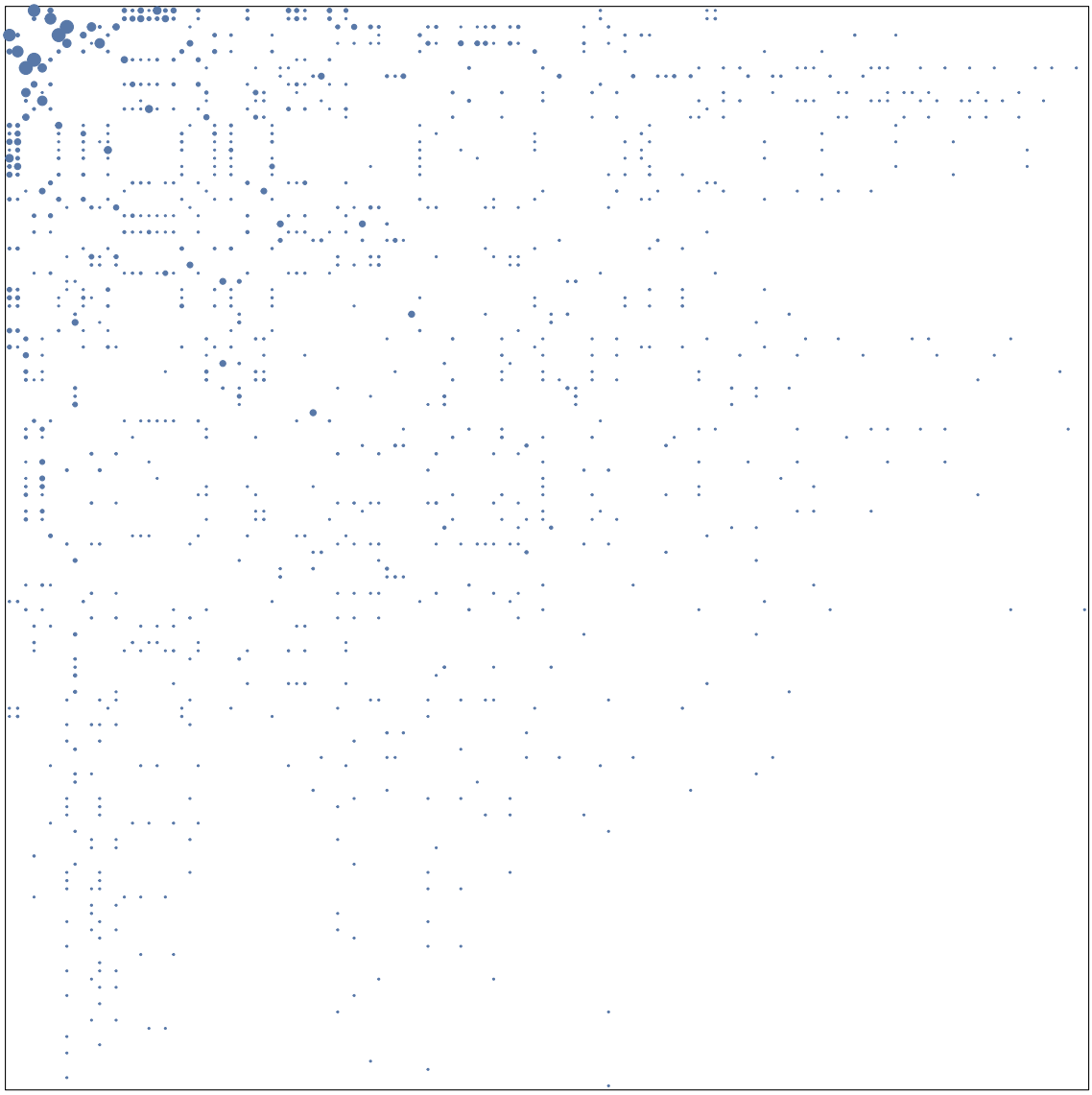
```r
# Dimension of the supramatrix:
#
# Here I count all the trees three times and all the animals three times.
# The interactions between animals in layers where they are not present are
# included in SAM as NAs.
SAM_dim<- (dim(pm.herb.mat)[1]*3)+ # Counts of trees
          (dim(pm.herb.mat)[2]*3)+
          (dim(pm.poll.mat)[2]*3)+
          (dim(pm.disp.mat)[2]*3)  ; SAM_dim
```

```
[1] 282
```

```r
visweb(pru.sam_bipartite$M, type= "none",
       circles=TRUE, boxes=F,
       labsize=1, circle.max=1, circle.min=0.15,
       text="no", outerbox.border="black", circle.col="#5878A8",
       box.col="lightgrey")
```



```r
visweb(pru.sam_bipartite$M,
       circles=TRUE, boxes=F,
       labsize=1, circle.max=1.5, circle.min=0.25,
       text="no", outerbox.border="black", circle.col="#5878A8",
       box.col="lightgrey")
```



```r
# These are the unipartite graphs.
g_layers_bip <- get_igraph(multilayer = multilayer_bip_pru, bipartite = T,
                           directed = F)
```

```
Joining with `by = join_by(node_name, layer_name)`
```

```
Joining with `by = join_by(node_name, node_id)`
```

```r
l1 <- g_layers_bip$layers_igraph[[1]]
l2 <- g_layers_bip$layers_igraph[[2]]
l3 <- g_layers_bip$layers_igraph[[3]]
l1; l2; l3
```

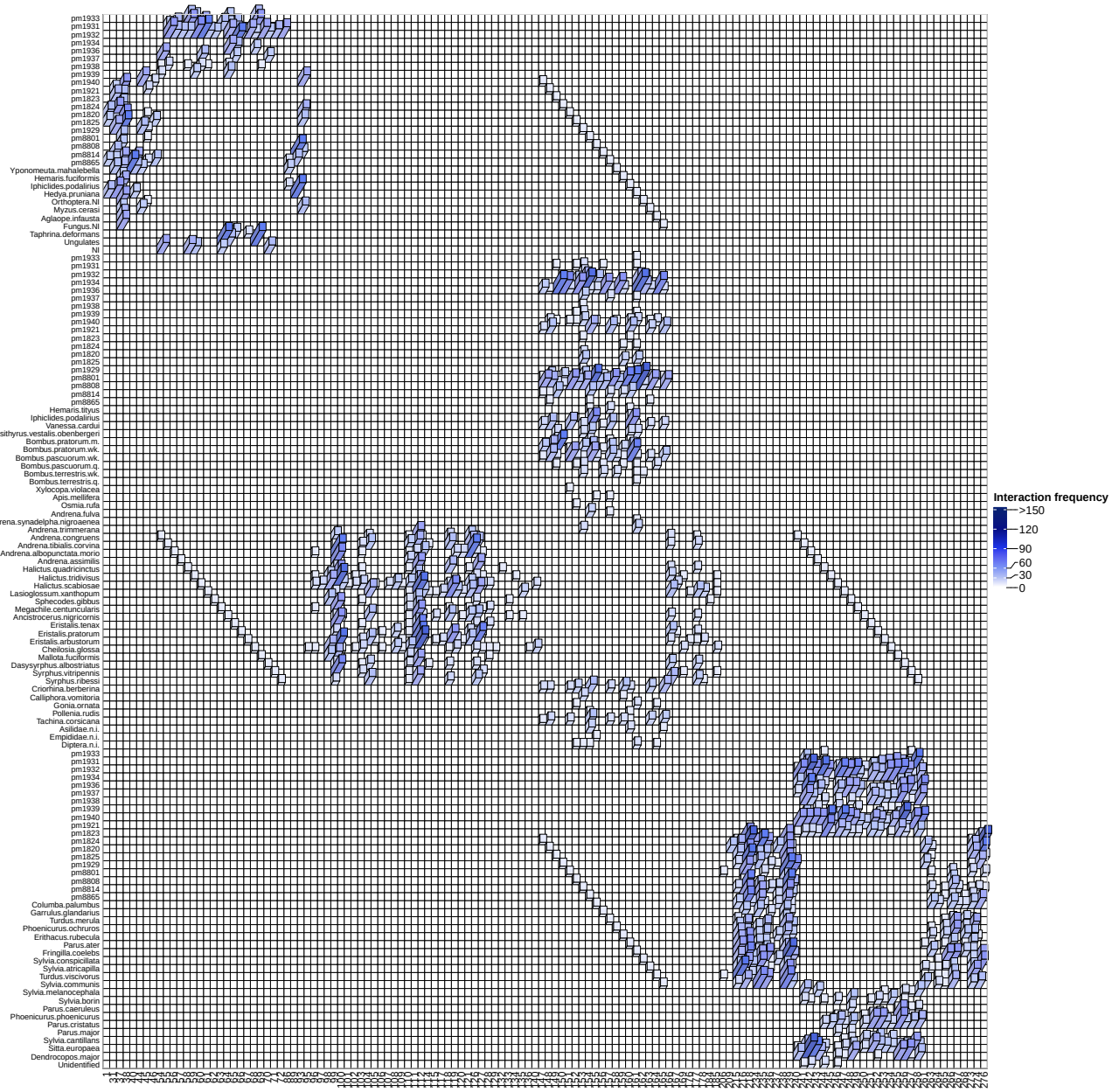
```
IGRAPH 2870d99 UNWB 30 59 --
+ attr: name (v/c), layer_id (v/n), node_id (v/n), type (v/l), weight
| (e/n)
+ edges from 2870d99 (vertex names):
 [1] Yponomeuta.mahalebella--pm1934 Yponomeuta.mahalebella--pm1921
 [3] Yponomeuta.mahalebella--pm1820 Yponomeuta.mahalebella--pm1929
 [5] Yponomeuta.mahalebella--pm8808 Hemaris.fuciformis    --pm1937
 [7] Iphiclides.podalirius --pm1936 Iphiclides.podalirius --pm1937
 [9] Iphiclides.podalirius --pm1940 Hedya.pruniana        --pm1933
```

```
[11] Hedya.pruniana        --pm1931 Hedya.pruniana        --pm1932
[13] Hedya.pruniana        --pm1936 Hedya.pruniana        --pm1937
+ ... omitted several edges

IGRAPH 34ae53d UNWB 62 219 --
+ attr: name (v/c), layer_id (v/n), node_id (v/n), type (v/l), weight
| (e/n)
+ edges from 34ae53d (vertex names):
 [1] Hemaris.tityus                --pm1921
 [2] Iphiclides.podalirius         --pm1933
 [3] Iphiclides.podalirius         --pm1936
 [4] Vanessa.cardui                --pm1933
 [5] Psithyrus.vestalis.obenbergeri--pm1931
 [6] Psithyrus.vestalis.obenbergeri--pm1938
 [7] Bombus.pratorum.m.            --pm1931
+ ... omitted several edges

IGRAPH 5c12746 UNWB 40 215 --
+ attr: name (v/c), layer_id (v/n), node_id (v/n), type (v/l), weight
| (e/n)
+ edges from 5c12746 (vertex names):
 [1] Columba.palumbus   --pm1921 Columba.palumbus   --pm8814
 [3] Garrulus.glandarius--pm1933 Garrulus.glandarius--pm1931
 [5] Garrulus.glandarius--pm1932 Garrulus.glandarius--pm1934
 [7] Garrulus.glandarius--pm1921 Garrulus.glandarius--pm1823
 [9] Garrulus.glandarius--pm1824 Garrulus.glandarius--pm1820
[11] Garrulus.glandarius--pm1825 Garrulus.glandarius--pm8801
[13] Garrulus.glandarius--pm8808 Garrulus.glandarius--pm8814
+ ... omitted several edges
```

These are 3D heatmap plots of the supra-adjacency matrix.

```r
# 3d heatmap
#
mat<- pru.sam_bipartite$M
rownames(mat)<- c("pm1933","pm1931","pm1932","pm1934","pm1936","pm1937",
                  "pm1938","pm1939","pm1940","pm1921","pm1823","pm1824",
                  "pm1820","pm1825","pm1929","pm8801","pm8808","pm8814",
                  "pm8865","Yponomeuta.mahalebella","Hemaris.fuciformis",
                  "Iphiclides.podalirius","Hedya.pruniana","Orthoptera.NI",
                  "Myzus.cerasi","Aglaope.infausta","Fungus.NI",
                  "Taphrina.deformans","Ungulates","NI","pm1933","pm1931",
                  "pm1932","pm1934","pm1936","pm1937","pm1938","pm1939",
                  "pm1940","pm1921","pm1823","pm1824","pm1820","pm1825",
                  "pm1929","pm8801","pm8808","pm8814","pm8865",
                  "Hemaris.tityus","Iphiclides.podalirius","Vanessa.cardui",
                  "Psithyrus.vestalis.obenbergeri","Bombus.pratorum.m.",
                  "Bombus.pratorum.wk.","Bombus.pascuorum.wk.",
                  "Bombus.pascuorum.q.","Bombus.terrestris.wk.",
                  "Bombus.terrestris.q.","Xylocopa.violacea","Apis.mellifera",
                  "Osmia.rufa","Andrena.fulva","Andrena.synadelpha.nigroaenea",
                  "Andrena.trimmerana","Andrena.congruens",
                  "Andrena.tibialis.corvina","Andrena.albopunctata.morio",
                  "Andrena.assimilis","Halictus.quadricinctus",
                  "Halictus.tridivisus","Halictus.scabiosae",
```

```
                    "Lasioglossum.xanthopum","Sphecodes.gibbus",
                    "Megachile.centuncularis","Ancistrocerus.nigricornis",
                    "Eristalis.tenax","Eristalis.pratorum","Eristalis.arbustorum",
                    "Cheilosia.glossa","Mallota.fuciformis",
                    "Dasysyrphus.albostriatus","Syrphus.vitripennis",
                    "Syrphus.ribessi","Criorhina.berberina","Calliphora.vomitoria",
                    "Gonia.ornata","Pollenia.rudis","Tachina.corsicana",
                    "Asilidae.n.i.","Empididae.n.i.","Diptera.n.i.","pm1933",
                    "pm1931","pm1932","pm1934","pm1936","pm1937","pm1938",
                    "pm1939","pm1940","pm1921","pm1823","pm1824","pm1820",
                    "pm1825","pm1929","pm8801","pm8808","pm8814","pm8865",
                    "Columba.palumbus","Garrulus.glandarius","Turdus.merula",
                    "Phoenicurus.ochruros","Erithacus.rubecula","Parus.ater",
                    "Fringilla.coelebs","Sylvia.conspicillata",
                    "Sylvia.atricapilla","Turdus.viscivorus","Sylvia.communis",
                    "Sylvia.melanocephala","Sylvia.borin","Parus.caeruleus",
                    "Phoenicurus.phoenicurus","Parus.cristatus","Parus.major",
                    "Sylvia.cantillans","Sitta.europaea","Dendrocopos.major",
                    "Unidentified")

# Plot
col_fun = circlize::colorRamp2(c(0, 1, 2, 4, 6, 8),
                               c("#FFFFFF", "#B3C0F5", "#7A93F5",
                                 "#1E36EB", "#070F85","#082173"))
ht_opt$TITLE_PADDING = unit(c(10, 2), "mm")

# This takes time...
Heatmap3D(log(mat+1), name = c("Interaction frequency"), col = col_fun,
          cluster_columns = FALSE, show_row_dend = FALSE,
          cluster_rows = FALSE,
          show_column_names = TRUE,
          heatmap_legend_param = list(at = c(0, 1, 2, 4, 6, 8),
                                      labels = c("0", "30", "60",
                                                 "90", "120",">150")),
          column_title = '      ',
          row_names_side = "left",
          column_names_gp = gpar(fontsize = 9),
          border = TRUE,
          border_gp = gpar(col = "lightgrey"),
          row_names_gp = gpar(fontsize = 7),
          # new arguments for Heatmap3D()
          bar_rel_width = 1, bar_rel_height = 1,
          bar_max_length = unit(0.7, "cm")
          )
```



Comparing EC scores in monolayer and multilayer networks

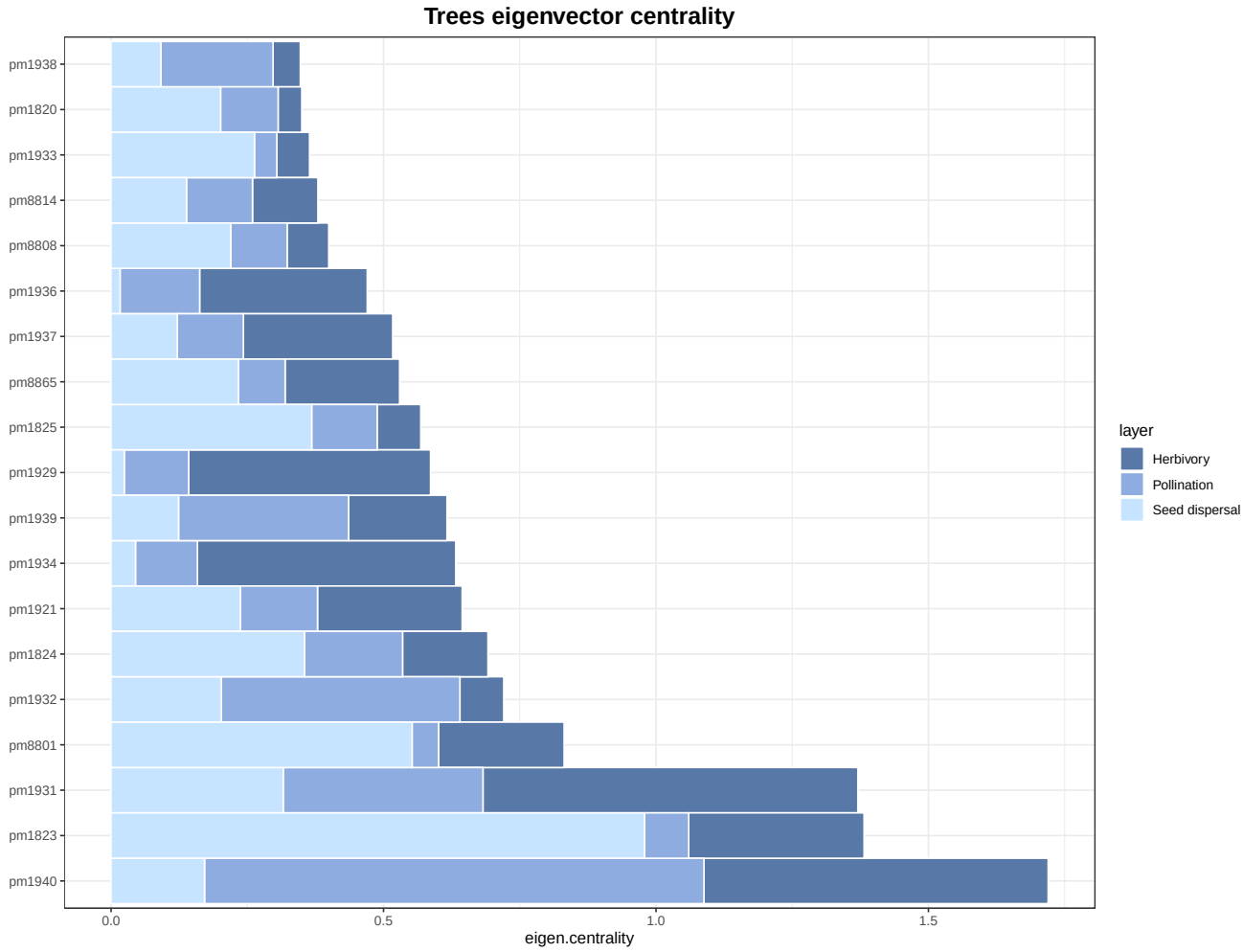
Calculating the Eigenvector centrality score of each state node on a per-layer basis. Then using a stack bar plot to include also the multilayer eigen-centralities.

```
# Get a list of igraph objects as layers igraph
g_layers_bip <- get_igraph(multilayer = multilayer_bip_pru,
                           bipartite = T, directed = F)
```

```
Joining with `by = join_by(node_name, layer_name)`
Joining with `by = join_by(node_name, node_id)`
```

```
# get Eigenvector scores of igraph layers
eigen_igraph_herb<-
    igraph::eigen_centrality(g_layers_bip$layers_igraph$'Herbivory')$vector
eigen_igraph_poll<-
    igraph::eigen_centrality(g_layers_bip$layers_igraph$'Pollination')$vector
eigen_igraph_disp<-
    igraph::eigen_centrality(g_layers_bip$layers_igraph$'Seed dispersal')$vector
```

```r
# Save the mono-layer eigenvector centralities
eigen_igraph_mono<- list(eigen_igraph_herb,eigen_igraph_poll,
                            eigen_igraph_disp)

eigenc_mono<- read.table(here::here("data/raw/multiplex/pm_eigen_centrality_mono.tsv"),
                         sep="\t", dec=".", header= TRUE)

# Stackbar plots for monolayer centralities of the trees.
# TODO
# Extract the trees
eigenc_mono_trees<- read.table(here::here(
    "data/raw/multiplex/pm_eigen_centrality_mono_trees.tsv"),
    sep="\t", dec=".", header= TRUE)

#-----------------------------------------------------------------
# Only trees eigenvector centrality.
eigenc_bar<- eigenc_mono_trees
colnames(eigenc_bar)<- c("layer","node","eigen.centrality")

# Plotting just the monolayer centralities, stacked.
ggplot(eigenc_bar, aes(x= reorder(node, -eigen.centrality),
                       y=eigen.centrality, fill=layer))+
  geom_bar(stat="identity", width=1, col="white")+
  theme_bw()+
  scale_x_discrete(name="")+
  ggtitle("Trees eigenvector centrality") +
  theme(plot.title = element_text(face = "bold", size = 16, colour = "black",
                                  hjust = 0.5, vjust = 1)) +
  scale_fill_manual(values=c("#5878A8", "#8EACDF", "#C6E4FF")) +
  coord_flip()
```




```r
#-----------------------------------------------------------------
# The multilayer
#-----------------------------------------------------------------
# My function for bipartite graph.
bip_init_igraph <- function (mat) {
                require(igraph)
                igraph::graph_from_incidence_matrix(mat,
                        weighted= T, add.names= T)
}
#-----------------------------------------------------------------
# get the SAM
pru.sam_bipartite <- get_sam(multilayer = multilayer_bip_pru, bipartite = T,
                             directed = F, sparse = F, remove_zero_rows_cols = T)
```

```
Joining with `by = join_by(node_name, layer_name)`
```

```r
# converting SAM to igraph object
graph_sam.g <- bip_init_igraph(pru.sam_bipartite$M)
```

```
Warning: `graph_from_incidence_matrix()` was deprecated in igraph 1.6.0.
i Please use `graph_from_biadjacency_matrix()` instead.
```

```r
graph_sam.g
```

```
IGRAPH e54ed0a U-WB 264 1062 --
+ attr: type (v/l), TRUE (v/c), weight (e/n)
+ edges from e54ed0a:
 [1]  1--145  1--146  1--151  1--155  2--142  2--143  2--145  2--146  2--147
```

```
[10]  2--150  2--151  2--152  2--155  3--142  3--143  3--144  3--145  3--146
[19]  3--147  3--148  3--149  3--151  3--152  3--153  3--154  3--155  3--156
[28]  3--157  3--158  3--159  4--152  5--151  5--152  5--155  6--141  6--147
[37]  6--152  6--157  7--141  7--142  7--145  7--147  7--148  7--151  7--156
[46]  8--141  8--146  8--151  9--138  9--139  9--140  9--162  9--198 10--134
[55] 10--135 10--139 10--199 11--134 11--135 11--200 12--135 12--201 13--133
[64] 13--134 13--135 13--139 13--162 13--202 14--133 14--134 14--135 14--140
+ ... omitted several edges
```

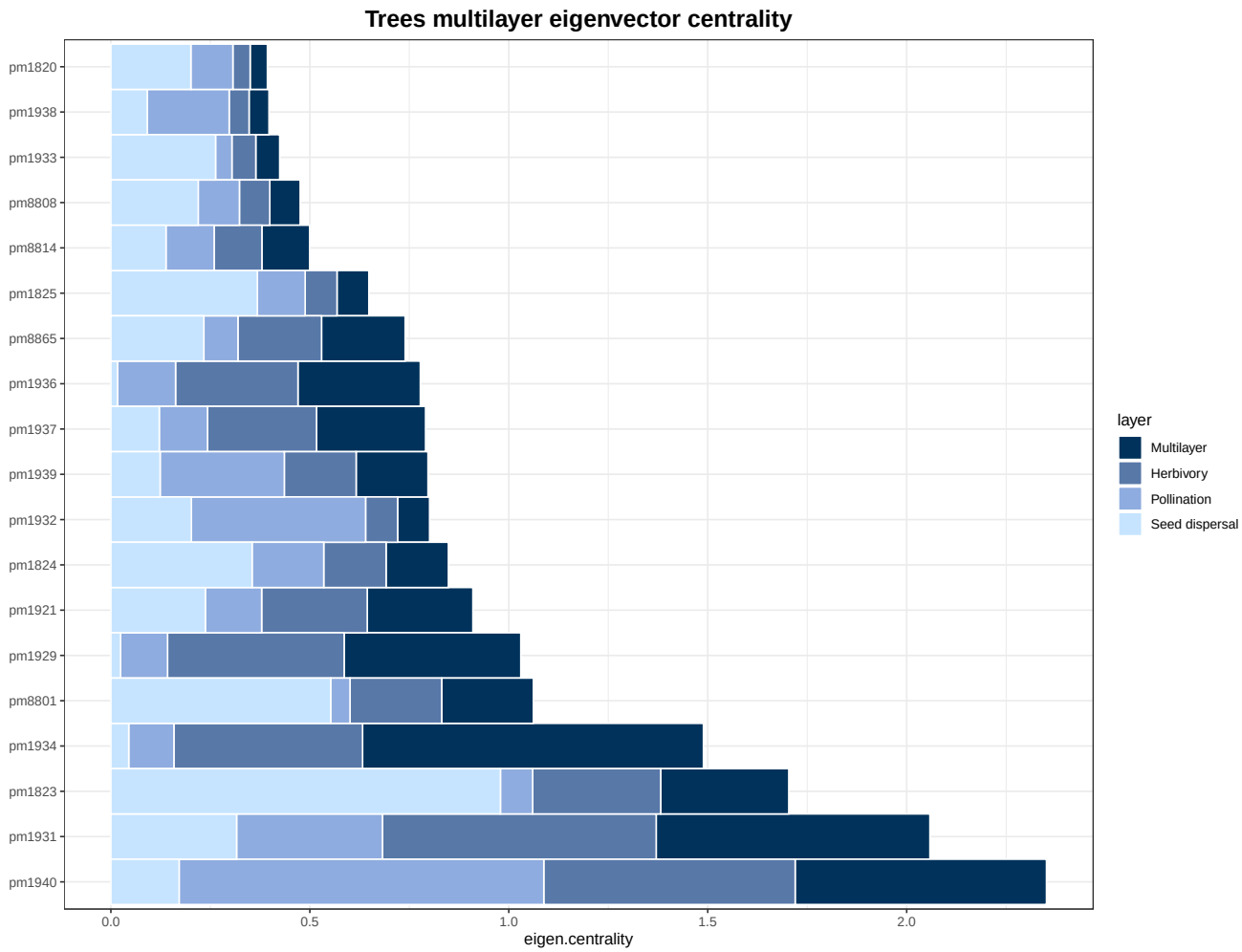
```
#-----------------------------------------------------------------
# Tree multi-layer eigenvector centrality
# eigenc_bar2<-as.data.frame(eigenc_bar2)

# Get Eigenvector scores
eigen_sam.g <- igraph::eigen_centrality(graph_sam.g, directed = F, scale = T)$vector
```

```
Warning: The `scale` argument of `eigen_centrality()` is deprecated as of igraph 2.1.1.
i eigen_centrality() will always behave as if scale=TRUE were used.
```

```
# Extract the trees
cond<- c(1:19,94:112,187:205)
eigen_SAM<- as.numeric(eigen_sam.g[cond])

# Merge to data frame eigenvector comparison
ec_comparison <- data.frame(eigen_SAM, eigenc_bar$eigen.centrality) ## CHECK!!!
colnames(ec_comparison) <- c("multilayer","monolayer")
ec_comparison$monolayer[4]  <- 0.8567376385295642 # Correct typo
                                                  # in tree pm1934.
                                                  # in tree pm1934.
eigen_igraph<- ec_comparison[,2]
eigen_multi_trees<- eigen_igraph[1:19]


eigen_multi_trees<- as.data.frame(eigen_multi_trees)
colnames(eigen_multi_trees)<- c("Multilayer")
eigen_multi_trees[4,]<- 0.8567376385295642   # Correct typo
                                             # in tree pm1934.

xx<- data.frame(cbind(rep("Multilayer",19),
                      eigenc_bar$node[1:19],
                      eigen_multi_trees$Multilayer))
colnames(xx)<- c("layer","node","eigen.centrality")

# Alter column formats
xx <- xx %>%
  mutate(
    layer = as.character(layer),
    node = as.character(node),
    eigen.centrality= as.numeric(eigen.centrality)
  )

eigen_centr <- xx %>%
bind_rows(eigenc_bar)
levels(as.factor(eigen_centr$layer))
```

```
[1] "Herbivory"      "Multilayer"     "Pollination"    "Seed dispersal"
```

```
eigen_centr$layer<- factor(eigen_centr$layer, levels=c('Multilayer',
                                                       'Herbivory',
                                                       'Pollination',
                                                       'Seed dispersal'))

ggplot(eigen_centr, aes(x= reorder(node, -eigen.centrality),
                        y= eigen.centrality, fill=layer))+
  geom_bar(stat="identity", width=1, col="white")+
  scale_fill_manual(values=c("#00325B", "#5878A8", "#8EACDF", "#C6E4FF")) +
  theme_bw()+
  scale_x_discrete(name="")+
  ggtitle("Trees multilayer eigenvector centrality") +
  theme(plot.title = element_text(face = "bold", size = 16, colour = "black",
                                  hjust = 0.5, vjust = 1)) +
  coord_flip()
```

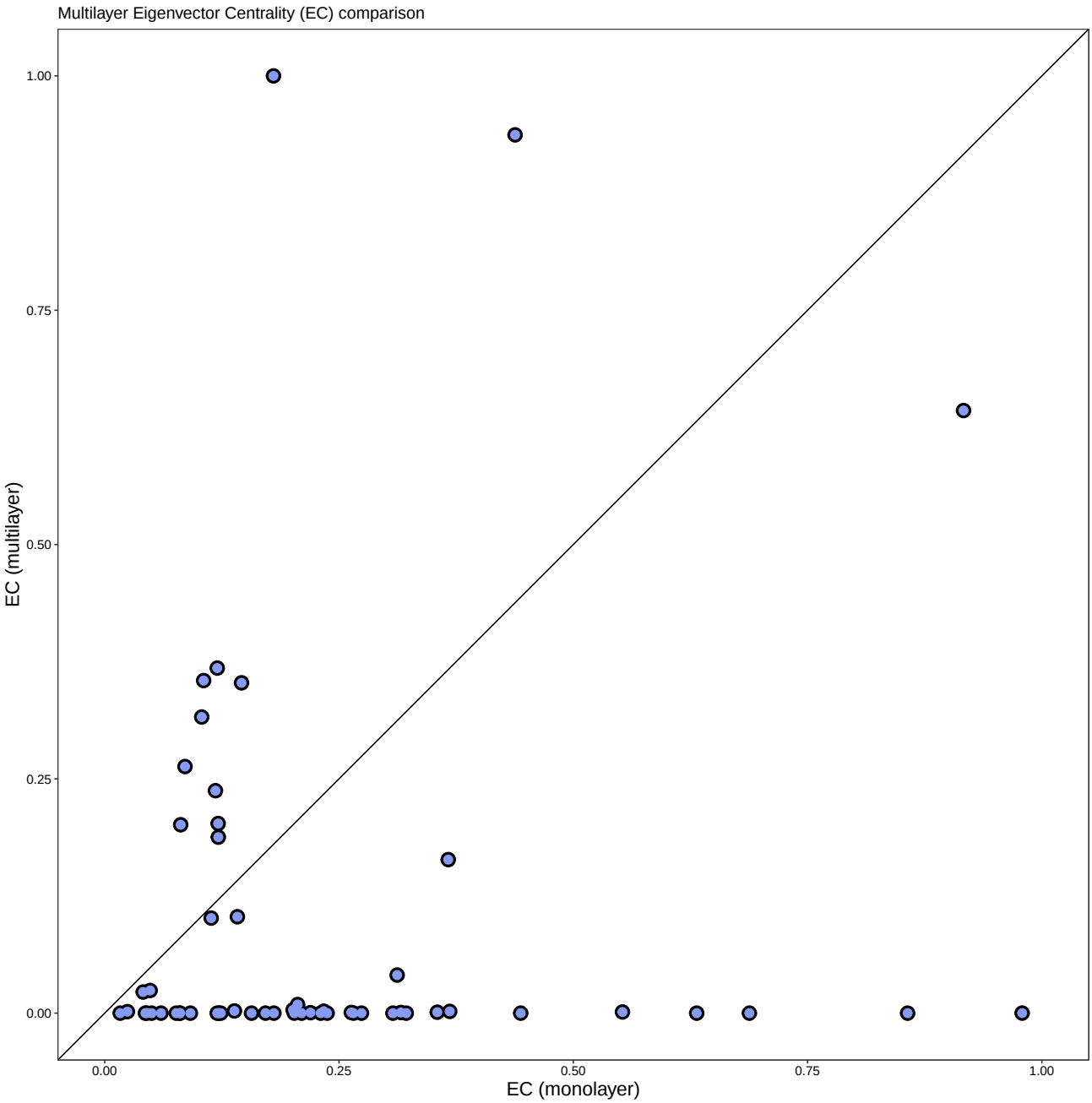



```
####################
```

Now calculating the eigenvector centrality score of each state node (the trees) using a SAM of the multilayer network. Then I compare the multilayer eigenv-centrality with the monolayer centralities.

```
# Scatter plot for eigenvector centralities comparison
ggplot(data = ec_comparison, aes(monolayer, multilayer)) +
            geom_point(aes(color= "#00325B"), size= 5, color= "black") +
            geom_point(color = "#859BF2", size = 3) +
```

```
          labs(title = "Multilayer Eigenvector Centrality (EC) comparison",
               x = "EC (monolayer)", y = "EC (multilayer)")+
  geom_abline()+
  scale_x_continuous(limits = c(0,1))+
  scale_y_continuous(limits = c(0,1))+
  coord_fixed() +
  theme_bw()+
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.title = element_text(size=15),
        axis.text = element_text(color='black',size = 10),
        legend.text =  element_text(size=15),
        legend.title = element_text(size=20))
```



The deviation of the points from the reference line indicates a lack of correlation between the two centrality scores. We see that the majority of the nodes that are important in a monolayer, lose this importance in the multilayer network. There is also a group of species that are important in both

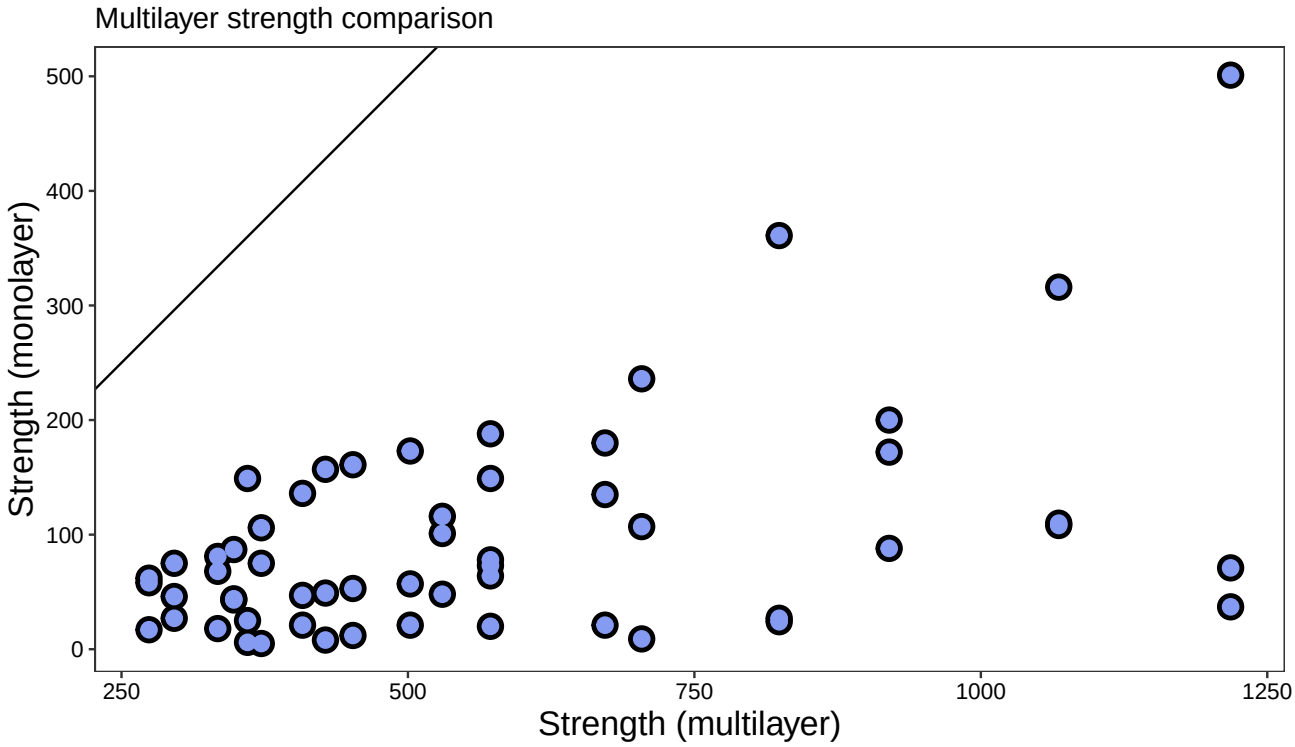
network types (upper right corner). This highlights the value of considering the unique ecological processes and interactions represented by different layers when analyzing multilayer networks and interpreting node importance.

Strength comparison

```
#|> echo= TRUE
#|> fig.height= 9
#|> fig.width= 12

# Data
strength<- read.table(here::here("RCode/multiplex/emln_input/strength.csv"),
                          sep=",", dec=".", header= TRUE)

# Stack the outcome columns
strength_df1 <- strength %>%
  dplyr::filter(node==c(1:19)) %>%
  pivot_longer(
    cols = c(layer1, layer2, layer3),
    names_to = "layer",
    values_to = "strength")
```

```
Warning: There was 1 warning in `dplyr::filter()`.
i In argument: `node == c(1:19)`.
Caused by warning in `node == c(1:19)`:
! longer object length is not a multiple of shorter object length
```

```
# Scatter plot for eigenvector centralities comparison
ggplot(data = strength_df1, aes(agg, strength)) +
            geom_point(aes(color= "#00325B"), size= 5, color= "black") +
            geom_point(color = "#859BF2", size = 3) +
              labs(title = "Multilayer strength comparison",
                x = "Strength (multilayer)", y = "Strength (monolayer)")+
  geom_abline()+
  coord_fixed() +
  theme_bw()+
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.title = element_text(size=15),
        axis.text = element_text(color='black',size = 10),
        legend.text =  element_text(size=15),
        legend.title = element_text(size=20))
```

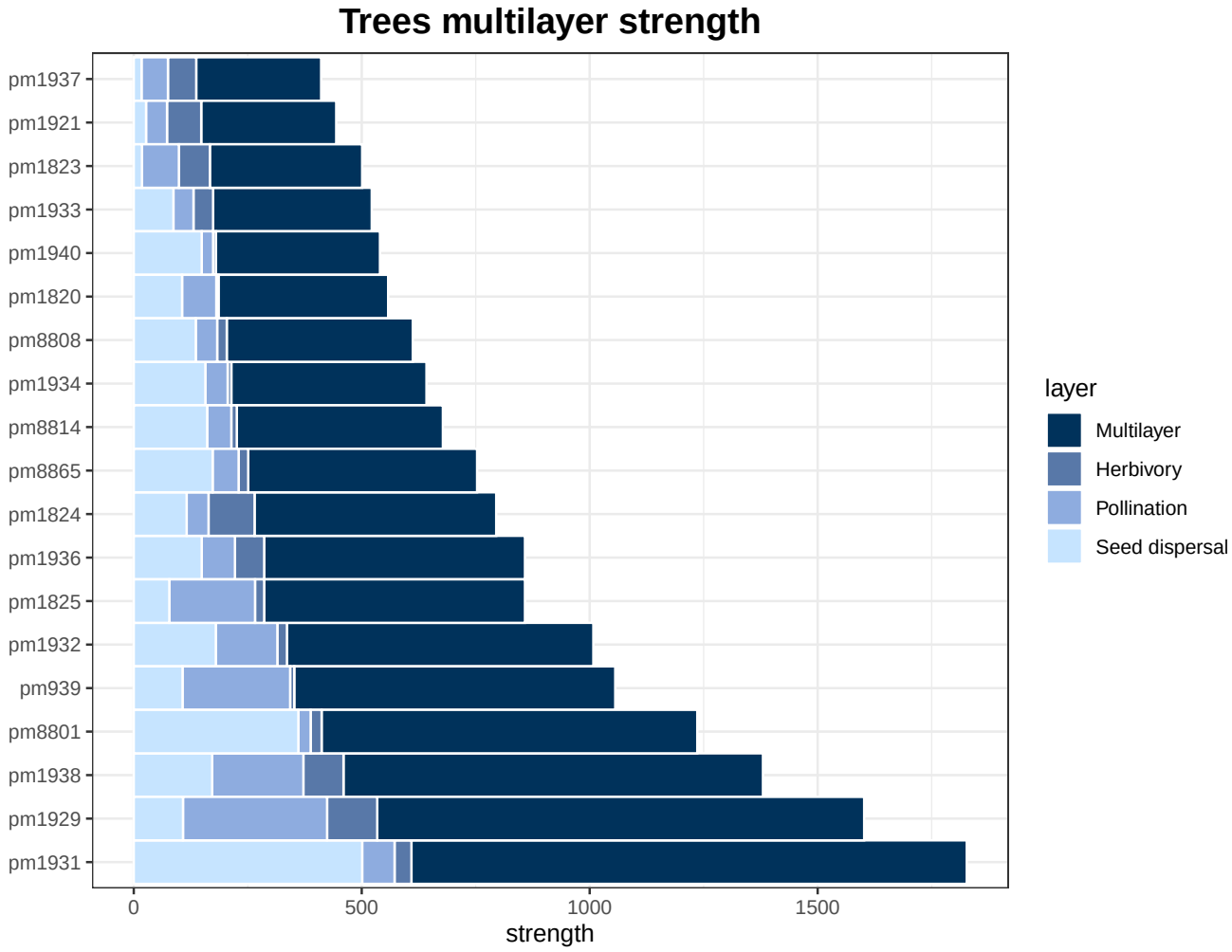



```
# Stack the outcome columns adding the data for multilayer strength.
strength_df2 <- strength %>%
  dplyr::filter(node==c(1:19)) %>%
  pivot_longer(
    cols = c(agg, layer1, layer2, layer3),
    names_to = "layer",
    values_to = "strength")
```

```
Warning: There was 1 warning in `dplyr::filter()`.
i In argument: `node == c(1:19)`.
Caused by warning in `node == c(1:19)`:
! longer object length is not a multiple of shorter object length
```

```
strength_df <- strength_df2 %>%
mutate(layer = recode(layer, 'agg'= 'Multilayer', 'layer1' = 'Herbivory',
                         'layer2' = 'Pollination',
                         'layer3' = 'Seed dispersal'))

strength_df$layer<- as.factor(strength_df$layer)
strength_df$layer<- factor(strength_df$layer, levels=c('Multilayer',
                                                       'Herbivory',
                                                       'Pollination',
                                                       'Seed dispersal'))

strength_df$tree<- rep(c("pm1933","pm1931","pm1932","pm1934","pm1936",
                                  "pm1937","pm1938","pm939","pm1940","pm1921",
                                  "pm1823","pm1824","pm1820","pm1825","pm1929",
                                  "pm8801","pm8808","pm8814","pm8865"), each = 4  )
ggplot(strength_df, aes(x= reorder(tree, -strength),
                        y= strength, fill=layer))+
  geom_bar(stat="identity", width=1, col="white")+
  scale_fill_manual(values=c("#00325B", "#5878A8", "#8EACDF", "#C6E4FF")) +
  theme_bw()+
```

```
  scale_x_discrete(name="")+
  ggtitle("Trees multilayer strength") +
  theme(plot.title = element_text(face = "bold", size = 16, colour = "black",
                                  hjust = 0.5, vjust = 1)) +
  coord_flip()
```



Modularity SAM with bipartite

Modularity with bipartite SAM network and with infomap.

```
# NOT RUN ##################################################################
#library(emln)
#install_infomap()
source(here::here("RCode/functions/my_run_infomap_multilayer.R"))
result <- my_run_infomap_multilayer(
    M = multilayer_bip_pru,
    infomap_executable = "Infomap", # pathto Infomap
    flow_model = "undirected", # Specify flow model explicitly
    silent = FALSE, # Enable verbose output for debugging
    trials = 100,
    relax = TRUE, # Disable relaxation if input format is extended
    run_standalone=T) # Run standalone execution
```

```
Warning: `as.tbl()` was deprecated in dplyr 1.0.0.
i Please use `tibble::as_tibble()` instead.
```

```
[1] "Using global relax to determine flow between layers."
[1] "./Infomap infomap_multilayer.txt . --tree -2 -N 100 -f undirected --multilayer-relax-ra
[1] "Removing auxilary files..."
[1] "Partitioned into 7 modules."
```

```r
# NOT RUN ######################################################################
#

# Testing modularity significance (@30 min for 100 null replicates)
require(bipartite)
TIME <- Sys.time()
mat <- pru.sam_bipartite$M     # Assign the observed matrix
node_names_supramux_132<-
    read.table(here::here("./data/raw/prunus_data/node_names_supramux_132.txt"))
rownames(mat)<- node_names_supramux_132[,1]

res <- computeModules(mat) # Modularity for observed matrix
nulls <- nullmodel(mat, N=5, method=3)
modules.nulls <- sapply(nulls, computeModules)
like.nulls <- sapply(modules.nulls, function(x) x@likelihood)
z <- (res@likelihood - mean(like.nulls))/sd(like.nulls)
p <- 2*pnorm(-abs(z))
cat("M= ", res@likelihood)
```

```
M=  0.74853
```

```r
cat("P= ",sprintf("%E", p))
```

```
P=  0.000000E+00
```

```r
Sys.time() - TIME
```

```
Time difference of 4.7315 mins
```

```r
printoutModuleInformation(res)
```

```


Depth: 0

Nr of module: 1

Rownames:

    Herb_1820
    Herb_1823
    Herb_1824
    Herb_1825
    Herb_1921
    Herb_1929
    Herb_1931
    Herb_1932
    Herb_1933
    Herb_1934
    Herb_1936
    Herb_1937
    Herb_1938
    Herb_1939
```

Herb_1940
Herb_8801
Herb_8808
Herb_8814
Herb_8865
Herb_Aglaope.infausta
Herb_Fungus.NI
Herb_Hedya.pruniana
Herb_Hemaris.fuciformis
Herb_Iphiclides.podalirius
Herb_Myzus.cerasi
Herb_NI
Herb_Orthoptera.NI
Herb_Taphrina.deformans
Herb_Ungulates
Herb_Yponomeuta.mahalebella
Poll_1820
Poll_1823
Poll_1824
Poll_1825
Poll_1921
Poll_1929
Poll_1931
Poll_1932
Poll_1933
Poll_1934
Poll_1936
Poll_1937
Poll_1938
Poll_1939
Poll_1940
Poll_8801
Poll_8808
Poll_8814
Poll_8865
Poll_Ancistrocerus.nigricornis
Poll_Andrena.albopunctata.morio
Poll_Andrena.assimilis
Poll_Andrena.congruens
Poll_Andrena.fulva
Poll_Andrena.synadelpha.nigroaenea
Poll_Andrena.tibialis.corvina
Poll_Andrena.trimmerana
Poll_Apis.mellifera
Poll_Asilidae.n.i.
Poll_Bombus.pascuorum.q.
Poll_Bombus.pascuorum.wk.
Poll_Bombus.pratorum.m.
Poll_Bombus.pratorum.wk.
Poll_Bombus.terrestris.q.
Poll_Bombus.terrestris.wk.
Poll_Calliphora.vomitoria
Poll_Cheilosia.glossa

Poll_Criorhina.berberina
Poll_Dasysyrphus.albostriatus
Poll_Diptera.n.i.
Poll_Empididae.n.i.
Poll_Eristalis.arbustorum
Poll_Eristalis.pratorum
Poll_Eristalis.tenax
Poll_Gonia.ornata
Poll_Halictus.quadricinctus
Poll_Halictus.scabiosae
Poll_Halictus.tridivisus
Poll_Hemaris.tityus
Poll_Iphiclides.podalirius
Poll_Lasioglossum.xanthopum
Poll_Mallota.fuciformis
Poll_Megachile.centuncularis
Poll_Osmia.rufa
Poll_Pollenia.rudis
Poll_Psithyrus.vestalis.obenbergeri
Poll_Sphecodes.gibbus
Poll_Syrphus.ribessi
Poll_Syrphus.vitripennis
Poll_Tachina.corsicana
Poll_Vanessa.cardui
Poll_Xylocopa.violacea
Disp_1820
Disp_1823
Disp_1824
Disp_1825
Disp_1921
Disp_1929
Disp_1931
Disp_1932
Disp_1933
Disp_1934
Disp_1936
Disp_1937
Disp_1938
Disp_1939
Disp_1940
Disp_8801
Disp_8808
Disp_8814
Disp_8865
Disp_Columba.palumbus
Disp_Dendrocopos.major
Disp_Erithacus.rubecula
Disp_Fringilla.coelebs
Disp_Garrulus.glandarius
Disp_Unidentified
Disp_Parus.ater
Disp_Parus.caeruleus
Disp_Parus.cristatus

```
    Disp_Parus.major
    Disp_Phoenicurus.ochruros
    Disp_Phoenicurus.phoenicurus
    Disp_Sitta.europaea
    Disp_Sylvia.atricapilla
    Disp_Sylvia.borin
    Disp_Sylvia.cantillans
    Disp_Sylvia.communis
    Disp_Sylvia.conspicillata
    Disp_Sylvia.melanocephala
    Disp_Turdus.merula
    Disp_Turdus.viscivorus

Colnames:

    1
    31
    37
    38
    40
    44
    45
    46
    54
    55
    56
    57
    58
    59
    60
    61
    62
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    64
    65
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    101
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184

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    185
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    257
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    261
    263
    264
    265
    266
    267
    268
    273
    274
    276

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Depth: 1

Nr of module: 1

Rownames:

    Disp_1937
```

```
    Disp_1938
    Disp_1939
    Disp_1940
    Disp_8801
    Disp_8808
    Disp_8814
    Disp_8865
    Disp_Columba.palumbus
    Disp_Dendrocopos.major
    Disp_Erithacus.rubecula
    Disp_Fringilla.coelebs
    Disp_Garrulus.glandarius
    Disp_Unidentified
    Disp_Parus.ater
    Disp_Parus.caeruleus
    Disp_Parus.cristatus
    Disp_Parus.major
    Disp_Phoenicurus.ochruros

Colnames:

    206
    209
    215
    216
    218
    234
    235
    236
    237
    238
    239
    261
    263
    264
    265
    266
    267
    268
    273
    274
    276

--------------------------

Depth: 1

Nr of module: 2

Rownames:

    Herb_1825
    Herb_1933
```

```
    Herb_1936
    Herb_1938
    Herb_8814
    Herb_Aglaope.infausta
    Herb_Hemaris.fuciformis
    Herb_Myzus.cerasi
    Herb_Taphrina.deformans

Colnames:

    1
    31
    38
    44
    65
    86
    89
    93


--------------------------

Depth: 1

Nr of module: 3

Rownames:

    Disp_1820
    Disp_1823
    Disp_1824
    Disp_1825
    Disp_1921
    Disp_1929
    Disp_1931
    Disp_1932
    Disp_1933
    Disp_1934
    Disp_1936
    Disp_Phoenicurus.phoenicurus
    Disp_Sitta.europaea
    Disp_Sylvia.atricapilla
    Disp_Sylvia.borin
    Disp_Sylvia.cantillans
    Disp_Sylvia.communis
    Disp_Sylvia.conspicillata
    Disp_Sylvia.melanocephala
    Disp_Turdus.merula
    Disp_Turdus.viscivorus

Colnames:

    240
    241
```

```
    242
    243
    244
    245
    246
    247
    248
    249
    250
    251
    252
    253
    254
    255
    256
    257
    258

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Depth: 1

Nr of module: 4

Rownames:

    Herb_1934
    Herb_1937
    Herb_1939
    Herb_1940
    Herb_8801
    Herb_8808
    Herb_8865
    Herb_Fungus.NI
    Herb_Hedya.pruniana
    Herb_Iphiclides.podalirius
    Herb_NI
    Herb_Orthoptera.NI

Colnames:

    37
    40
    45
    46

--------------------------

Depth: 1

Nr of module: 5

Rownames:
```

```
    Poll_1823
    Poll_1824
    Poll_1825
    Poll_1921
    Poll_1929
    Poll_1931
    Poll_1932
    Poll_1933
    Poll_1936
    Poll_1938
    Poll_8801
    Poll_Andrena.synadelpha.nigroaenea
    Poll_Andrena.trimmerana
    Poll_Apis.mellifera
    Poll_Bombus.pascuorum.wk.
    Poll_Bombus.pratorum.m.
    Poll_Psithyrus.vestalis.obenbergeri
    Poll_Sphecodes.gibbus
    Poll_Vanessa.cardui
    Poll_Xylocopa.violacea

Colnames:

    149
    150
    152
    153
    155
    156
    158
    162
    164

--------------------------

Depth: 1

Nr of module: 6

Rownames:

    Herb_1820
    Herb_1823
    Herb_1824
    Herb_1921
    Herb_1931
    Herb_1932
    Herb_Ungulates
    Herb_Yponomeuta.mahalebella
    Poll_1820
    Poll_1934
    Poll_1937
```

```
    Poll_1939
    Poll_1940
    Poll_8808
    Poll_8814
    Poll_8865
    Poll_Ancistrocerus.nigricornis
    Poll_Andrena.albopunctata.morio
    Poll_Andrena.assimilis
    Poll_Andrena.congruens
    Poll_Andrena.fulva
    Poll_Andrena.tibialis.corvina
    Poll_Asilidae.n.i.
    Poll_Bombus.pascuorum.q.
    Poll_Bombus.pratorum.wk.
    Poll_Bombus.terrestris.q.
    Poll_Bombus.terrestris.wk.
    Poll_Pollenia.rudis
    Poll_Syrphus.ribessi
    Poll_Syrphus.vitripennis
    Poll_Tachina.corsicana

Colnames:

    55
    56
    57
    58
    59
    60
    61
    62
    63
    64
    66
    67
    68
    69
    70
    71
    72
    147
    148
    151
    154
    157
    159
    160
    161
    163
    165

--------------------------
```

```
Depth: 1

Nr of module: 7

Rownames:

    Herb_1929

Colnames:

    54

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Depth: 1

Nr of module: 8

Rownames:

    Poll_Calliphora.vomitoria
    Poll_Cheilosia.glossa
    Poll_Criorhina.berberina
    Poll_Dasysyrphus.albostriatus
    Poll_Diptera.n.i.
    Poll_Empididae.n.i.
    Poll_Eristalis.arbustorum
    Poll_Eristalis.pratorum
    Poll_Eristalis.tenax
    Poll_Gonia.ornata
    Poll_Halictus.quadricinctus
    Poll_Halictus.scabiosae
    Poll_Halictus.tridivisus
    Poll_Hemaris.tityus
    Poll_Iphiclides.podalirius
    Poll_Lasioglossum.xanthopum
    Poll_Mallota.fuciformis
    Poll_Megachile.centuncularis
    Poll_Osmia.rufa

Colnames:

    95
    96
    97
    98
    99
    100
    101
    102
    103
    104
    105
```

```
    106
    107
    108
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    140
    166
    167
    169
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    178
    184
    185

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--------------------------
```
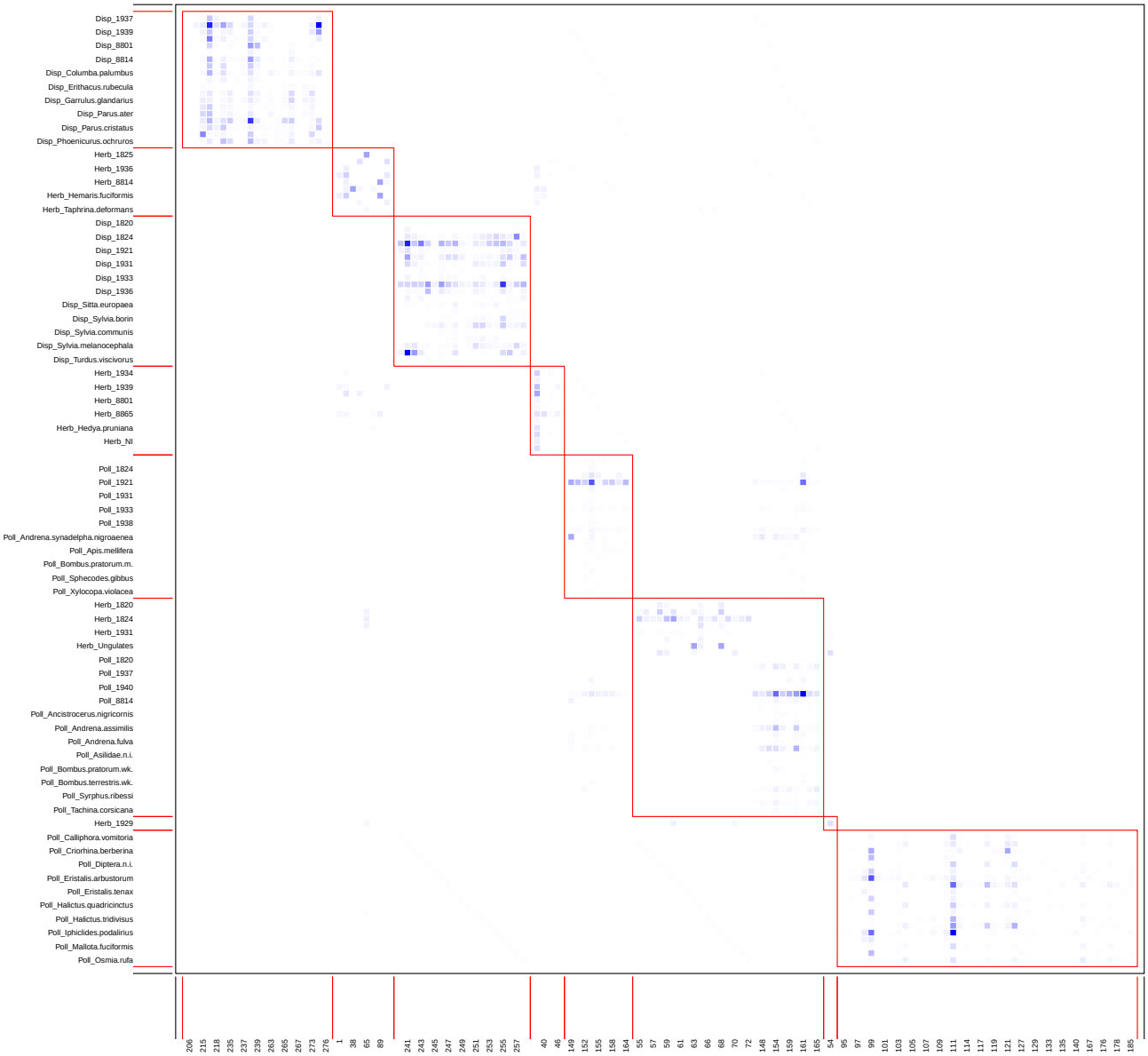

```
plotModuleWeb(res, plotModules = TRUE,
rank = FALSE, weighted = TRUE, displayAlabels = TRUE,
displayBlabels = TRUE, labsize = 0.5, xlabel = "", ylabel = "",
square.border = "white", fromDepth = 0, upToDepth = -1)
```




```r
# Print the whole network to assess node names
pru.sam_bipartite %>% print(n=Inf, na.print = "<NA>")
```

```
$M
      1 31 37 38 40 44 45 46 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70
1     0  0  0  0  0  0  0  0  0  0  0  0 10  5  0  0  0  0  5  0  0  0 11  0  0
31    0  0  0  0  0  0  0  0  0  4 11  0 30  3 17  0  0 10  5  8  0  0 30  0  0
37    0  0  0  0  0  0  0  0  0 30 10  8  8 41 59  8  6  0 18 11  5 18  4 23  6
38    0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 62  0  0  0  0  0
40    0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0 18 10  0  0  5  0  0
44    0  0  0  0  0  0  0  0 20  0  0  0  0  0 10  0  0  0  0  5  0  0  0  0  5
45    0  0  0  0  0  0  0  0  1  3  0  0  1  0  2  1  0  0  2  0  0  0  0  1  0
46    0  0  0  0  0  0  0  0  2  0  0  0  0  5  0  0  0  0  8  0  0  0  0  0  0
54    0  0  0  0  0 20  1  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
55    0  4 30  0  0  0  3  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
56    0 11 10  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
57    0  0  8  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
58   10 30  8  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
59    5  3 41  0  0  0  0  5  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
60    0 17 59  0  0 10  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
```

61	0	0	8	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
62	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
63	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
64	5	5	18	0	18	0	2	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
65	0	8	11	62	10	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
66	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
67	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
68	11	30	4	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
69	0	0	23	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
70	0	0	6	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
71	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
72	0	0	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
86	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	5	0	2	0	0	0
89	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	60	10	0	0	0	60	0	0
93	0	0	0	0	0	0	0	0	20	0	0	0	15	8	0	0	0	5	0	0	0	0	0	0	10
95	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
96	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
97	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
98	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
99	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
100	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
101	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
102	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
103	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
104	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
105	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
106	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
107	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
108	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
109	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
110	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
111	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
112	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
114	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
115	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
117	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
118	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
119	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
120	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
121	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
126	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
127	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
128	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
132	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
133	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
134	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
135	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
136	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
140	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
147	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
148	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
149	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

150	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
151	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
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154	98	7	1	1	3	41	6	2	10	21	0	0	0	0	0	0	4	0	0
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161	166	0	0	0	0	1	3	0	0	2	1	1	1	0	0	0	0	1	4
162	3	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
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103	0	2	1	0	1	1	4	0	1	0	0	1	0	2	3	0	1	2	0
104	2	7	0	0	0	0	0	15	0	0	10	0	0	9	0	0	4	0	9

105	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0
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107	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0
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110	1	5	1	4	1	0	6	1	2	2	1	1	1	0	8	3	2	1	1
111	22	15	3	2	29	8	16	98	8	10	32	8	49	66	166	3	23	1	18
112	1	1	7	0	0	0	0	7	0	0	2	0	2	6	0	0	0	0	0
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119	0	0	3	0	0	0	2	6	0	0	0	0	0	3	3	0	0	0	0
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166	2	2	0	0	2	4	0	10	0	0	3	0	7	2	0	0	5	0	12
167	0	0	0	0	0	0	3	0	0	0	0	0	1	0	0	0	0	0	0
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	166	167	169	176	177	178	184	185	206	209	215	216	218	234	235	236	237	238	239
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126	0	0	0	0	0	0	0	0	0	0
127	0	0	0	0	0	0	0	0	0	0
128	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0
132	0	0	0	0	0	0	0	0	0	0
133	0	0	0	0	0	0	0	0	0	0
134	0	0	0	0	0	0	0	0	0	0
135	0	0	0	0	0	0	0	0	0	0
136	0	0	0	0	0	0	0	0	0	0
140	0	0	0	0	0	0	0	0	0	0

147	0	0	0	0	0	0	0	0	0	0
148	0	0	0	0	0	0	0	0	0	0
149	0	0	0	0	0	0	0	0	0	0
150	0	0	0	0	0	0	0	0	0	0
151	0	0	0	0	0	0	0	0	0	0
152	0	0	0	0	0	0	0	0	0	0
153	0	0	0	0	0	0	0	0	0	0
154	0	0	0	0	0	0	0	0	0	0
155	0	0	0	0	0	0	0	0	0	0
156	0	0	0	0	0	0	0	0	0	0
157	0	0	0	0	0	0	0	0	0	0
158	0	0	0	0	0	0	0	0	0	0
159	0	0	0	0	0	0	0	0	0	0
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161	0	0	0	0	0	0	0	0	0	0
162	0	0	0	0	0	0	0	0	0	0
163	0	0	0	0	0	0	0	0	0	0
164	0	0	0	0	0	0	0	0	0	0
165	0	0	0	0	0	0	0	0	0	0
166	0	0	0	0	0	0	0	0	0	0
167	0	0	0	0	0	0	0	0	0	0
169	0	0	0	0	0	0	0	0	0	0
176	0	0	0	0	0	0	0	0	0	0
177	0	0	0	0	0	0	0	0	0	0
178	0	0	0	0	0	0	0	0	0	0
184	0	0	0	0	0	0	0	0	0	0
185	0	0	0	0	0	0	0	0	0	0
206	0	0	0	0	0	0	0	0	0	0
209	0	0	0	0	0	0	0	0	0	0
215	0	0	0	0	0	0	0	0	0	0
216	0	0	0	0	0	0	0	0	0	0
218	0	0	0	0	0	0	0	0	0	0
234	0	0	0	0	0	0	0	0	0	0
235	0	0	0	0	0	0	0	0	0	0
236	0	0	0	0	0	0	0	0	0	0
237	0	0	0	0	0	0	0	0	0	0
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239	0	0	0	0	0	0	0	0	0	0
240	0	0	0	0	0	0	0	1	10	0
241	9	2	0	0	0	0	0	11	168	1
242	0	1	0	0	0	0	0	16	66	2
243	5	0	0	0	0	0	0	1	13	0
244	0	1	0	0	3	0	0	3	0	2
245	0	0	0	0	5	0	0	0	0	0
246	0	2	0	1	11	2	0	1	2	1
247	0	0	0	1	2	0	1	3	1	0
248	0	7	3	0	7	2	2	6	18	0
249	0	0	0	0	0	0	0	0	0	0
250	0	0	0	1	3	0	4	2	0	0
251	0	1	0	3	25	0	3	20	0	0
252	0	3	0	4	27	0	7	8	1	0
253	1	1	0	1	6	0	1	6	0	0
254	0	6	0	0	3	0	4	4	0	0

```
255    0    0    1   27   32    0    0    4   24    0
256    2    1    1    2    5    0    0    5   34    0
257    1    0    1    1    8    0    0   25    1    0
258    7    0    2    1   13    0    0    5    9    0
261    0    0    0    0    0    0    0    0    0    0
263    0    0    0    0    0    0    0    0    0    0
264    0    0    0    0    0    0    0    0    0    0
265    0    0    0    0    0    0    0    0    0    0
266    0    0    0    0    0    0    0    0    0    0
267    0    0    0    0    0    0    0    0    0    0
268    0    0    0    0    0    0    0    0    0    0
273    0    0    0    0    0    0    0    0    0    0
274    0    0    0    0    0    0    0    0    0    0
276    0    0    0    0    0    0    0    0    0    0

$nodes
# A tibble: 93 x 2
   node_id node_name
     <int> <chr>
 1       1 Aglaope.infausta
 2       2 Ancistrocerus.nigricornis
 3       3 Andrena.albopunctata.morio
 4       4 Andrena.assimilis
 5       5 Andrena.congruens
 6       6 Andrena.fulva
 7       7 Andrena.synadelpha.nigroaenea
 8       8 Andrena.tibialis.corvina
 9       9 Andrena.trimmerana
10      10 Apis.mellifera
11      11 Asilidae.n.i.
12      12 Bombus.pascuorum.q.
13      13 Bombus.pascuorum.wk.
14      14 Bombus.pratorum.m.
15      15 Bombus.pratorum.wk.
16      16 Bombus.terrestris.q.
17      17 Bombus.terrestris.wk.
18      18 Calliphora.vomitoria
19      19 Cheilosia.glossa
20      20 Columba.palumbus
21      21 Criorhina.berberina
22      22 Dasysyrphus.albostriatus
23      23 Dendrocopos.major
24      24 Diptera.n.i.
25      25 Empididae.n.i.
26      26 Eristalis.arbustorum
27      27 Eristalis.pratorum
28      28 Eristalis.tenax
29      29 Erithacus.rubecula
30      30 Fringilla.coelebs
31      31 Fungus.NI
32      32 Garrulus.glandarius
33      33 Gonia.ornata
34      34 Halictus.quadricinctus
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35      35 Halictus.scabiosae
36      36 Halictus.tridivisus
37      37 Hedya.pruniana
38      38 Hemaris.fuciformis
39      39 Hemaris.tityus
40      40 Iphiclides.podalirius
41      41 Lasioglossum.xanthopum
42      42 Mallota.fuciformis
43      43 Megachile.centuncularis
44      44 Myzus.cerasi
45      45 NI
46      46 Orthoptera.NI
47      47 Osmia.rufa
48      48 Parus.ater
49      49 Parus.caeruleus
50      50 Parus.cristatus
51      51 Parus.major
52      52 Phoenicurus.ochruros
53      53 Phoenicurus.phoenicurus
54      54 pm1820
55      55 pm1823
56      56 pm1824
57      57 pm1825
58      58 pm1921
59      59 pm1929
60      60 pm1931
61      61 pm1932
62      62 pm1933
63      63 pm1934
64      64 pm1936
65      65 pm1937
66      66 pm1938
67      67 pm1939
68      68 pm1940
69      69 pm8801
70      70 pm8808
71      71 pm8814
72      72 pm8865
73      73 Pollenia.rudis
74      74 Psithyrus.vestalis.obenbergeri
75      75 Sitta.europaea
76      76 Sphecodes.gibbus
77      77 Sylvia.atricapilla
78      78 Sylvia.borin
79      79 Sylvia.cantillans
80      80 Sylvia.communis
81      81 Sylvia.conspicillata
82      82 Sylvia.melanocephala
83      83 Syrphus.ribessi
84      84 Syrphus.vitripennis
85      85 Tachina.corsicana
86      86 Taphrina.deformans
87      87 Turdus.merula
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88      88 Turdus.viscivorus
89      89 Ungulates
90      90 Unidentified
91      91 Vanessa.cardui
92      92 Xylocopa.violacea
93      93 Yponomeuta.mahalebella

$state_nodes_map
    sn_id     layer_name                      node_name layer_id node_id
1       1      Herbivory               Aglaope.infausta        1       1
2       2      Herbivory      Ancistrocerus.nigricornis       NA      NA
3       3      Herbivory     Andrena.albopunctata.morio       NA      NA
4       4      Herbivory              Andrena.assimilis       NA      NA
5       5      Herbivory              Andrena.congruens       NA      NA
6       6      Herbivory                  Andrena.fulva       NA      NA
7       7      Herbivory  Andrena.synadelpha.nigroaenea       NA      NA
8       8      Herbivory       Andrena.tibialis.corvina       NA      NA
9       9      Herbivory             Andrena.trimmerana       NA      NA
10     10      Herbivory                 Apis.mellifera       NA      NA
11     11      Herbivory                  Asilidae.n.i.       NA      NA
12     12      Herbivory            Bombus.pascuorum.q.       NA      NA
13     13      Herbivory           Bombus.pascuorum.wk.       NA      NA
14     14      Herbivory             Bombus.pratorum.m.       NA      NA
15     15      Herbivory            Bombus.pratorum.wk.       NA      NA
16     16      Herbivory           Bombus.terrestris.q.       NA      NA
17     17      Herbivory          Bombus.terrestris.wk.       NA      NA
18     18      Herbivory           Calliphora.vomitoria       NA      NA
19     19      Herbivory               Cheilosia.glossa       NA      NA
20     20      Herbivory               Columba.palumbus       NA      NA
21     21      Herbivory            Criorhina.berberina       NA      NA
22     22      Herbivory       Dasysyrphus.albostriatus       NA      NA
23     23      Herbivory              Dendrocopos.major       NA      NA
24     24      Herbivory                   Diptera.n.i.       NA      NA
25     25      Herbivory                 Empididae.n.i.       NA      NA
26     26      Herbivory           Eristalis.arbustorum       NA      NA
27     27      Herbivory             Eristalis.pratorum       NA      NA
28     28      Herbivory                Eristalis.tenax       NA      NA
29     29      Herbivory             Erithacus.rubecula       NA      NA
30     30      Herbivory              Fringilla.coelebs       NA      NA
31     31      Herbivory                       Fungus.NI        1      31
32     32      Herbivory            Garrulus.glandarius       NA      NA
33     33      Herbivory                   Gonia.ornata       NA      NA
34     34      Herbivory         Halictus.quadricinctus       NA      NA
35     35      Herbivory             Halictus.scabiosae       NA      NA
36     36      Herbivory            Halictus.tridivisus       NA      NA
37     37      Herbivory                 Hedya.pruniana        1      37
38     38      Herbivory             Hemaris.fuciformis        1      38
39     39      Herbivory                 Hemaris.tityus       NA      NA
40     40      Herbivory          Iphiclides.podalirius        1      40
41     41      Herbivory         Lasioglossum.xanthopum       NA      NA
42     42      Herbivory             Mallota.fuciformis       NA      NA
43     43      Herbivory        Megachile.centuncularis       NA      NA
44     44      Herbivory                   Myzus.cerasi        1      44
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45	45	Herbivory	NI	1	45
46	46	Herbivory	Orthoptera.NI	1	46
47	47	Herbivory	Osmia.rufa	NA	NA
48	48	Herbivory	Parus.ater	NA	NA
49	49	Herbivory	Parus.caeruleus	NA	NA
50	50	Herbivory	Parus.cristatus	NA	NA
51	51	Herbivory	Parus.major	NA	NA
52	52	Herbivory	Phoenicurus.ochruros	NA	NA
53	53	Herbivory	Phoenicurus.phoenicurus	NA	NA
54	54	Herbivory	pm1820	1	54
55	55	Herbivory	pm1823	1	55
56	56	Herbivory	pm1824	1	56
57	57	Herbivory	pm1825	1	57
58	58	Herbivory	pm1921	1	58
59	59	Herbivory	pm1929	1	59
60	60	Herbivory	pm1931	1	60
61	61	Herbivory	pm1932	1	61
62	62	Herbivory	pm1933	1	62
63	63	Herbivory	pm1934	1	63
64	64	Herbivory	pm1936	1	64
65	65	Herbivory	pm1937	1	65
66	66	Herbivory	pm1938	1	66
67	67	Herbivory	pm1939	1	67
68	68	Herbivory	pm1940	1	68
69	69	Herbivory	pm8801	1	69
70	70	Herbivory	pm8808	1	70
71	71	Herbivory	pm8814	1	71
72	72	Herbivory	pm8865	1	72
73	73	Herbivory	Pollenia.rudis	NA	NA
74	74	Herbivory	Psithyrus.vestalis.obenbergeri	NA	NA
75	75	Herbivory	Sitta.europaea	NA	NA
76	76	Herbivory	Sphecodes.gibbus	NA	NA
77	77	Herbivory	Sylvia.atricapilla	NA	NA
78	78	Herbivory	Sylvia.borin	NA	NA
79	79	Herbivory	Sylvia.cantillans	NA	NA
80	80	Herbivory	Sylvia.communis	NA	NA
81	81	Herbivory	Sylvia.conspicillata	NA	NA
82	82	Herbivory	Sylvia.melanocephala	NA	NA
83	83	Herbivory	Syrphus.ribessi	NA	NA
84	84	Herbivory	Syrphus.vitripennis	NA	NA
85	85	Herbivory	Tachina.corsicana	NA	NA
86	86	Herbivory	Taphrina.deformans	1	86
87	87	Herbivory	Turdus.merula	NA	NA
88	88	Herbivory	Turdus.viscivorus	NA	NA
89	89	Herbivory	Ungulates	1	89
90	90	Herbivory	Unidentified	NA	NA
91	91	Herbivory	Vanessa.cardui	NA	NA
92	92	Herbivory	Xylocopa.violacea	NA	NA
93	93	Herbivory	Yponomeuta.mahalebella	1	93
94	94	Pollination	Aglaope.infausta	NA	NA
95	95	Pollination	Ancistrocerus.nigricornis	2	2
96	96	Pollination	Andrena.albopunctata.morio	2	3
97	97	Pollination	Andrena.assimilis	2	4

98	98	Pollination	Andrena.congruens	2	5
99	99	Pollination	Andrena.fulva	2	6
100	100	Pollination	Andrena.synadelpha.nigroaenea	2	7
101	101	Pollination	Andrena.tibialis.corvina	2	8
102	102	Pollination	Andrena.trimmerana	2	9
103	103	Pollination	Apis.mellifera	2	10
104	104	Pollination	Asilidae.n.i.	2	11
105	105	Pollination	Bombus.pascuorum.q.	2	12
106	106	Pollination	Bombus.pascuorum.wk.	2	13
107	107	Pollination	Bombus.pratorum.m.	2	14
108	108	Pollination	Bombus.pratorum.wk.	2	15
109	109	Pollination	Bombus.terrestris.q.	2	16
110	110	Pollination	Bombus.terrestris.wk.	2	17
111	111	Pollination	Calliphora.vomitoria	2	18
112	112	Pollination	Cheilosia.glossa	2	19
113	113	Pollination	Columba.palumbus	NA	NA
114	114	Pollination	Criorhina.berberina	2	21
115	115	Pollination	Dasysyrphus.albostriatus	2	22
116	116	Pollination	Dendrocopos.major	NA	NA
117	117	Pollination	Diptera.n.i.	2	24
118	118	Pollination	Empididae.n.i.	2	25
119	119	Pollination	Eristalis.arbustorum	2	26
120	120	Pollination	Eristalis.pratorum	2	27
121	121	Pollination	Eristalis.tenax	2	28
122	122	Pollination	Erithacus.rubecula	NA	NA
123	123	Pollination	Fringilla.coelebs	NA	NA
124	124	Pollination	Fungus.NI	NA	NA
125	125	Pollination	Garrulus.glandarius	NA	NA
126	126	Pollination	Gonia.ornata	2	33
127	127	Pollination	Halictus.quadricinctus	2	34
128	128	Pollination	Halictus.scabiosae	2	35
129	129	Pollination	Halictus.tridivisus	2	36
130	130	Pollination	Hedya.pruniana	NA	NA
131	131	Pollination	Hemaris.fuciformis	NA	NA
132	132	Pollination	Hemaris.tityus	2	39
133	133	Pollination	Iphiclides.podalirius	2	40
134	134	Pollination	Lasioglossum.xanthopum	2	41
135	135	Pollination	Mallota.fuciformis	2	42
136	136	Pollination	Megachile.centuncularis	2	43
137	137	Pollination	Myzus.cerasi	NA	NA
138	138	Pollination	NI	NA	NA
139	139	Pollination	Orthoptera.NI	NA	NA
140	140	Pollination	Osmia.rufa	2	47
141	141	Pollination	Parus.ater	NA	NA
142	142	Pollination	Parus.caeruleus	NA	NA
143	143	Pollination	Parus.cristatus	NA	NA
144	144	Pollination	Parus.major	NA	NA
145	145	Pollination	Phoenicurus.ochruros	NA	NA
146	146	Pollination	Phoenicurus.phoenicurus	NA	NA
147	147	Pollination	pm1820	2	54
148	148	Pollination	pm1823	2	55
149	149	Pollination	pm1824	2	56
150	150	Pollination	pm1825	2	57

151	151	Pollination	pm1921	2	58
152	152	Pollination	pm1929	2	59
153	153	Pollination	pm1931	2	60
154	154	Pollination	pm1932	2	61
155	155	Pollination	pm1933	2	62
156	156	Pollination	pm1934	2	63
157	157	Pollination	pm1936	2	64
158	158	Pollination	pm1937	2	65
159	159	Pollination	pm1938	2	66
160	160	Pollination	pm1939	2	67
161	161	Pollination	pm1940	2	68
162	162	Pollination	pm8801	2	69
163	163	Pollination	pm8808	2	70
164	164	Pollination	pm8814	2	71
165	165	Pollination	pm8865	2	72
166	166	Pollination	Pollenia.rudis	2	73
167	167	Pollination	Psithyrus.vestalis.obenbergeri	2	74
168	168	Pollination	Sitta.europaea	NA	NA
169	169	Pollination	Sphecodes.gibbus	2	76
170	170	Pollination	Sylvia.atricapilla	NA	NA
171	171	Pollination	Sylvia.borin	NA	NA
172	172	Pollination	Sylvia.cantillans	NA	NA
173	173	Pollination	Sylvia.communis	NA	NA
174	174	Pollination	Sylvia.conspicillata	NA	NA
175	175	Pollination	Sylvia.melanocephala	NA	NA
176	176	Pollination	Syrphus.ribessi	2	83
177	177	Pollination	Syrphus.vitripennis	2	84
178	178	Pollination	Tachina.corsicana	2	85
179	179	Pollination	Taphrina.deformans	NA	NA
180	180	Pollination	Turdus.merula	NA	NA
181	181	Pollination	Turdus.viscivorus	NA	NA
182	182	Pollination	Ungulates	NA	NA
183	183	Pollination	Unidentified	NA	NA
184	184	Pollination	Vanessa.cardui	2	91
185	185	Pollination	Xylocopa.violacea	2	92
186	186	Pollination	Yponomeuta.mahalebella	NA	NA
187	187	Seed dispersal	Aglaope.infausta	NA	NA
188	188	Seed dispersal	Ancistrocerus.nigricornis	NA	NA
189	189	Seed dispersal	Andrena.albopunctata.morio	NA	NA
190	190	Seed dispersal	Andrena.assimilis	NA	NA
191	191	Seed dispersal	Andrena.congruens	NA	NA
192	192	Seed dispersal	Andrena.fulva	NA	NA
193	193	Seed dispersal	Andrena.synadelpha.nigroaenea	NA	NA
194	194	Seed dispersal	Andrena.tibialis.corvina	NA	NA
195	195	Seed dispersal	Andrena.trimmerana	NA	NA
196	196	Seed dispersal	Apis.mellifera	NA	NA
197	197	Seed dispersal	Asilidae.n.i.	NA	NA
198	198	Seed dispersal	Bombus.pascuorum.q.	NA	NA
199	199	Seed dispersal	Bombus.pascuorum.wk.	NA	NA
200	200	Seed dispersal	Bombus.pratorum.m.	NA	NA
201	201	Seed dispersal	Bombus.pratorum.wk.	NA	NA
202	202	Seed dispersal	Bombus.terrestris.q.	NA	NA
203	203	Seed dispersal	Bombus.terrestris.wk.	NA	NA

204	204	Seed dispersal	Calliphora.vomitoria	NA	NA
205	205	Seed dispersal	Cheilosia.glossa	NA	NA
206	206	Seed dispersal	Columba.palumbus	3	20
207	207	Seed dispersal	Criorhina.berberina	NA	NA
208	208	Seed dispersal	Dasysyrphus.albostriatus	NA	NA
209	209	Seed dispersal	Dendrocopos.major	3	23
210	210	Seed dispersal	Diptera.n.i.	NA	NA
211	211	Seed dispersal	Empididae.n.i.	NA	NA
212	212	Seed dispersal	Eristalis.arbustorum	NA	NA
213	213	Seed dispersal	Eristalis.pratorum	NA	NA
214	214	Seed dispersal	Eristalis.tenax	NA	NA
215	215	Seed dispersal	Erithacus.rubecula	3	29
216	216	Seed dispersal	Fringilla.coelebs	3	30
217	217	Seed dispersal	Fungus.NI	NA	NA
218	218	Seed dispersal	Garrulus.glandarius	3	32
219	219	Seed dispersal	Gonia.ornata	NA	NA
220	220	Seed dispersal	Halictus.quadricinctus	NA	NA
221	221	Seed dispersal	Halictus.scabiosae	NA	NA
222	222	Seed dispersal	Halictus.tridivisus	NA	NA
223	223	Seed dispersal	Hedya.pruniana	NA	NA
224	224	Seed dispersal	Hemaris.fuciformis	NA	NA
225	225	Seed dispersal	Hemaris.tityus	NA	NA
226	226	Seed dispersal	Iphiclides.podalirius	NA	NA
227	227	Seed dispersal	Lasioglossum.xanthopum	NA	NA
228	228	Seed dispersal	Mallota.fuciformis	NA	NA
229	229	Seed dispersal	Megachile.centuncularis	NA	NA
230	230	Seed dispersal	Myzus.cerasi	NA	NA
231	231	Seed dispersal	NI	NA	NA
232	232	Seed dispersal	Orthoptera.NI	NA	NA
233	233	Seed dispersal	Osmia.rufa	NA	NA
234	234	Seed dispersal	Parus.ater	3	48
235	235	Seed dispersal	Parus.caeruleus	3	49
236	236	Seed dispersal	Parus.cristatus	3	50
237	237	Seed dispersal	Parus.major	3	51
238	238	Seed dispersal	Phoenicurus.ochruros	3	52
239	239	Seed dispersal	Phoenicurus.phoenicurus	3	53
240	240	Seed dispersal	pm1820	3	54
241	241	Seed dispersal	pm1823	3	55
242	242	Seed dispersal	pm1824	3	56
243	243	Seed dispersal	pm1825	3	57
244	244	Seed dispersal	pm1921	3	58
245	245	Seed dispersal	pm1929	3	59
246	246	Seed dispersal	pm1931	3	60
247	247	Seed dispersal	pm1932	3	61
248	248	Seed dispersal	pm1933	3	62
249	249	Seed dispersal	pm1934	3	63
250	250	Seed dispersal	pm1936	3	64
251	251	Seed dispersal	pm1937	3	65
252	252	Seed dispersal	pm1938	3	66
253	253	Seed dispersal	pm1939	3	67
254	254	Seed dispersal	pm1940	3	68
255	255	Seed dispersal	pm8801	3	69
256	256	Seed dispersal	pm8808	3	70

257	257	Seed dispersal	pm8814	3	71
258	258	Seed dispersal	pm8865	3	72
259	259	Seed dispersal	Pollenia.rudis	NA	NA
260	260	Seed dispersal	Psithyrus.vestalis.obenbergeri	NA	NA
261	261	Seed dispersal	Sitta.europaea	3	75
262	262	Seed dispersal	Sphecodes.gibbus	NA	NA
263	263	Seed dispersal	Sylvia.atricapilla	3	77
264	264	Seed dispersal	Sylvia.borin	3	78
265	265	Seed dispersal	Sylvia.cantillans	3	79
266	266	Seed dispersal	Sylvia.communis	3	80
267	267	Seed dispersal	Sylvia.conspicillata	3	81
268	268	Seed dispersal	Sylvia.melanocephala	3	82
269	269	Seed dispersal	Syrphus.ribessi	NA	NA
270	270	Seed dispersal	Syrphus.vitripennis	NA	NA
271	271	Seed dispersal	Tachina.corsicana	NA	NA
272	272	Seed dispersal	Taphrina.deformans	NA	NA
273	273	Seed dispersal	Turdus.merula	3	87
274	274	Seed dispersal	Turdus.viscivorus	3	88
275	275	Seed dispersal	Ungulates	NA	NA
276	276	Seed dispersal	Unidentified	3	90
277	277	Seed dispersal	Vanessa.cardui	NA	NA
278	278	Seed dispersal	Xylocopa.violacea	NA	NA
279	279	Seed dispersal	Yponomeuta.mahalebella	NA	NA

	tuple
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2	Herbivory_Ancistrocerus.nigricornis
3	Herbivory_Andrena.albopunctata.morio
4	Herbivory_Andrena.assimilis
5	Herbivory_Andrena.congruens
6	Herbivory_Andrena.fulva
7	Herbivory_Andrena.synadelpha.nigroaenea
8	Herbivory_Andrena.tibialis.corvina
9	Herbivory_Andrena.trimmerana
10	Herbivory_Apis.mellifera
11	Herbivory_Asilidae.n.i.
12	Herbivory_Bombus.pascuorum.q.
13	Herbivory_Bombus.pascuorum.wk.
14	Herbivory_Bombus.pratorum.m.
15	Herbivory_Bombus.pratorum.wk.
16	Herbivory_Bombus.terrestris.q.
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18	Herbivory_Calliphora.vomitoria
19	Herbivory_Cheilosia.glossa
20	Herbivory_Columba.palumbus
21	Herbivory_Criorhina.berberina
22	Herbivory_Dasysyrphus.albostriatus
23	Herbivory_Dendrocopos.major
24	Herbivory_Diptera.n.i.
25	Herbivory_Empididae.n.i.
26	Herbivory_Eristalis.arbustorum
27	Herbivory_Eristalis.pratorum
28	Herbivory_Eristalis.tenax
29	Herbivory_Erithacus.rubecula

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30                    Herbivory_Fringilla.coelebs
31                            Herbivory_Fungus.NI
32                 Herbivory_Garrulus.glandarius
33                        Herbivory_Gonia.ornata
34              Herbivory_Halictus.quadricinctus
35                  Herbivory_Halictus.scabiosae
36                 Herbivory_Halictus.tridivisus
37                      Herbivory_Hedya.pruniana
38                  Herbivory_Hemaris.fuciformis
39                      Herbivory_Hemaris.tityus
40                Herbivory_Iphiclides.podalirius
41               Herbivory_Lasioglossum.xanthopum
42                  Herbivory_Mallota.fuciformis
43             Herbivory_Megachile.centuncularis
44                        Herbivory_Myzus.cerasi
45                                   Herbivory_NI
46                       Herbivory_Orthoptera.NI
47                          Herbivory_Osmia.rufa
48                          Herbivory_Parus.ater
49                      Herbivory_Parus.caeruleus
50                      Herbivory_Parus.cristatus
51                          Herbivory_Parus.major
52                 Herbivory_Phoenicurus.ochruros
53              Herbivory_Phoenicurus.phoenicurus
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55                                Herbivory_pm1823
56                                Herbivory_pm1824
57                                Herbivory_pm1825
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62                                Herbivory_pm1933
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66                                Herbivory_pm1938
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68                                Herbivory_pm1940
69                                Herbivory_pm8801
70                                Herbivory_pm8808
71                                Herbivory_pm8814
72                                Herbivory_pm8865
73                      Herbivory_Pollenia.rudis
74       Herbivory_Psithyrus.vestalis.obenbergeri
75                      Herbivory_Sitta.europaea
76                    Herbivory_Sphecodes.gibbus
77                  Herbivory_Sylvia.atricapilla
78                        Herbivory_Sylvia.borin
79                   Herbivory_Sylvia.cantillans
80                     Herbivory_Sylvia.communis
81                  Herbivory_Sylvia.conspicillata
82                  Herbivory_Sylvia.melanocephala
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83                    Herbivory_Syrphus.ribessi
84                Herbivory_Syrphus.vitripennis
85                  Herbivory_Tachina.corsicana
86                 Herbivory_Taphrina.deformans
87                      Herbivory_Turdus.merula
88                 Herbivory_Turdus.viscivorus
89                          Herbivory_Ungulates
90                       Herbivory_Unidentified
91                     Herbivory_Vanessa.cardui
92                  Herbivory_Xylocopa.violacea
93             Herbivory_Yponomeuta.mahalebella
94                 Pollination_Aglaope.infausta
95        Pollination_Ancistrocerus.nigricornis
96       Pollination_Andrena.albopunctata.morio
97                Pollination_Andrena.assimilis
98                Pollination_Andrena.congruens
99                    Pollination_Andrena.fulva
100   Pollination_Andrena.synadelpha.nigroaenea
101        Pollination_Andrena.tibialis.corvina
102              Pollination_Andrena.trimmerana
103                  Pollination_Apis.mellifera
104                   Pollination_Asilidae.n.i.
105             Pollination_Bombus.pascuorum.q.
106            Pollination_Bombus.pascuorum.wk.
107              Pollination_Bombus.pratorum.m.
108             Pollination_Bombus.pratorum.wk.
109            Pollination_Bombus.terrestris.q.
110           Pollination_Bombus.terrestris.wk.
111            Pollination_Calliphora.vomitoria
112                Pollination_Cheilosia.glossa
113                Pollination_Columba.palumbus
114             Pollination_Criorhina.berberina
115        Pollination_Dasysyrphus.albostriatus
116               Pollination_Dendrocopos.major
117                     Pollination_Diptera.n.i.
118                  Pollination_Empididae.n.i.
119            Pollination_Eristalis.arbustorum
120              Pollination_Eristalis.pratorum
121                 Pollination_Eristalis.tenax
122              Pollination_Erithacus.rubecula
123               Pollination_Fringilla.coelebs
124                        Pollination_Fungus.NI
125             Pollination_Garrulus.glandarius
126                    Pollination_Gonia.ornata
127          Pollination_Halictus.quadricinctus
128              Pollination_Halictus.scabiosae
129             Pollination_Halictus.tridivisus
130                  Pollination_Hedya.pruniana
131              Pollination_Hemaris.fuciformis
132                  Pollination_Hemaris.tityus
133           Pollination_Iphiclides.podalirius
134          Pollination_Lasioglossum.xanthopum
135              Pollination_Mallota.fuciformis
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136          Pollination_Megachile.centuncularis
137                     Pollination_Myzus.cerasi
138                               Pollination_NI
139                    Pollination_Orthoptera.NI
140                       Pollination_Osmia.rufa
141                       Pollination_Parus.ater
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144                      Pollination_Parus.major
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147                            Pollination_pm1820
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159                            Pollination_pm1938
160                            Pollination_pm1939
161                            Pollination_pm1940
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163                            Pollination_pm8808
164                            Pollination_pm8814
165                            Pollination_pm8865
166                   Pollination_Pollenia.rudis
167   Pollination_Psithyrus.vestalis.obenbergeri
168                   Pollination_Sitta.europaea
169                 Pollination_Sphecodes.gibbus
170               Pollination_Sylvia.atricapilla
171                     Pollination_Sylvia.borin
172                Pollination_Sylvia.cantillans
173                  Pollination_Sylvia.communis
174             Pollination_Sylvia.conspicillata
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178                 Pollination_Tachina.corsicana
179                Pollination_Taphrina.deformans
180                     Pollination_Turdus.merula
181                 Pollination_Turdus.viscivorus
182                         Pollination_Ungulates
183                     Pollination_Unidentified
184                   Pollination_Vanessa.cardui
185                 Pollination_Xylocopa.violacea
186            Pollination_Yponomeuta.mahalebella
187               Seed dispersal_Aglaope.infausta
188     Seed dispersal_Ancistrocerus.nigricornis
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189     Seed dispersal_Andrena.albopunctata.morio
190              Seed dispersal_Andrena.assimilis
191              Seed dispersal_Andrena.congruens
192                  Seed dispersal_Andrena.fulva
193 Seed dispersal_Andrena.synadelpha.nigroaenea
194      Seed dispersal_Andrena.tibialis.corvina
195             Seed dispersal_Andrena.trimmerana
196                 Seed dispersal_Apis.mellifera
197                  Seed dispersal_Asilidae.n.i.
198           Seed dispersal_Bombus.pascuorum.q.
199          Seed dispersal_Bombus.pascuorum.wk.
200            Seed dispersal_Bombus.pratorum.m.
201           Seed dispersal_Bombus.pratorum.wk.
202          Seed dispersal_Bombus.terrestris.q.
203         Seed dispersal_Bombus.terrestris.wk.
204          Seed dispersal_Calliphora.vomitoria
205              Seed dispersal_Cheilosia.glossa
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207           Seed dispersal_Criorhina.berberina
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210                  Seed dispersal_Diptera.n.i.
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214               Seed dispersal_Eristalis.tenax
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217                     Seed dispersal_Fungus.NI
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231                             Seed dispersal_NI
232                 Seed dispersal_Orthoptera.NI
233                    Seed dispersal_Osmia.rufa
234                    Seed dispersal_Parus.ater
235               Seed dispersal_Parus.caeruleus
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237                   Seed dispersal_Parus.major
238          Seed dispersal_Phoenicurus.ochruros
239       Seed dispersal_Phoenicurus.phoenicurus
240                        Seed dispersal_pm1820
241                        Seed dispersal_pm1823
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                        Seed dispersal_pm1824
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Seed dispersal_Psithyrus.vestalis.obenbergeri
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              Seed dispersal_Sylvia.cantillans
                Seed dispersal_Sylvia.communis
           Seed dispersal_Sylvia.conspicillata
           Seed dispersal_Sylvia.melanocephala
                Seed dispersal_Syrphus.ribessi
            Seed dispersal_Syrphus.vitripennis
              Seed dispersal_Tachina.corsicana
             Seed dispersal_Taphrina.deformans
                  Seed dispersal_Turdus.merula
              Seed dispersal_Turdus.viscivorus
                      Seed dispersal_Ungulates
                   Seed dispersal_Unidentified
                 Seed dispersal_Vanessa.cardui
              Seed dispersal_Xylocopa.violacea
         Seed dispersal_Yponomeuta.mahalebella
```

Software used

Package	Version	Citation
attempt	0.3.1	Fay (2020)
base	4.4.3	R Core Team (2025)
bipartite	2.21	(**biparti?**)….
ComplexHeatmap	2.22.0	(**Complex?**)….
cowplot	1.1.3	Wilke (2024)
devtools	2.4.5	Wickham et al. (2022)
DT	0.33	Xie, Cheng, and Tan (2024)
emln	1.0.2	Frydman, Pilosof, and Freilikhman (2025)
ggalluvial	0.12.5	(**ggalluv?**)….
ggbipart	0.1.2	Jordano (2025)
ggbreak	0.1.4	Shuangbin Xu et al. (2021)
ggpubr	0.6.0	Kassambara (2023)
ggsci	3.2.0	Xiao (2024)
hablar	0.3.2	Sjoberg (2023)
here	1.0.1	Müller (2020)
igraph	2.1.4	(**igraph2?**)….
infomapecology	2.0	(**infomap?**)….
knitr	1.50	(**knitr20?**)….
lattice	0.22.7	Sarkar (2008)
Matrix	1.7.3	Bates, Maechler, and Jagan (2025)
network	1.19.0	Butts (2015)….
permute	0.9.7	Simpson (2022)
sna	2.8	Butts (2024)
statnet.common	4.11.0	(**statnet?**)….
tidyverse	2.0.0	Wickham et al. (2019)
usethis	3.1.0	Wickham et al. (2024)
vegan	2.6.10	Oksanen et al. (2025)

Session info

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R version 4.4.3 (2025-02-28)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.4

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;

locale:
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time zone: Europe/Madrid
tzcode source: internal

attached base packages:
[1] grid      stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] infomapecology_2.0    rlang_1.1.5           ggalluvial_0.12.5
 [4] cowplot_1.1.3         attempt_0.3.1         ComplexHeatmap_2.22.0
 [7] grateful_0.2.11       ggbipart_0.1.2        ggpubr_0.6.0
[10] ggbreak_0.1.4         ggsci_3.2.0           bipartite_2.21
[13] vegan_2.6-10          lattice_0.22-7        permute_0.9-7
[16] sna_2.8               network_1.19.0        statnet.common_4.11.0
[19] emln_1.0.2            hablar_0.3.2          DT_0.33
[22] Matrix_1.7-3          igraph_2.1.4          magrittr_2.0.3
[25] knitr_1.50            devtools_2.4.5        usethis_3.1.0
[28] lubridate_1.9.4       forcats_1.0.0         stringr_1.5.1
[31] dplyr_1.1.4           purrr_1.0.4           readr_2.1.5
[34] tidyr_1.3.1           tibble_3.2.1          ggplot2_3.5.1
[37] tidyverse_2.0.0       here_1.0.1

loaded via a namespace (and not attached):
  [1] RColorBrewer_1.1-3  shape_1.4.6.1       rstudioapi_0.17.1
  [4] jsonlite_2.0.0      magick_2.8.6        farver_2.1.2
  [7] rmarkdown_2.29      GlobalOptions_0.1.2 fs_1.6.5
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 [34] S4Vectors_0.44.0    rprojroot_2.0.4     pkgload_1.4.0
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 [43] bit64_4.6.0-1       withr_3.0.2         doParallel_1.0.17
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 [64] utf8_1.2.4          car_3.1-3           BiocGenerics_0.52.0
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